Deloitte.

`

NATIONAL CARDIOVASCULAR DISEASE PREVENTION STRATEGY FOR QATAR 2019-2022

مستقبلنا صحتنا

Contents

1 SCOPE 4

1.1 Aim, Rationale and definitions 4
1.2 Risk Factors 4
1.3 Strategic Context 5
1.4 Burden of Disease 6

1.4.1 Mortality and premature mortality 6
1.4.2 Prevalence 7
1.4.3 Demographic factors 8
1.4.4 Patients with undiagnosed risk factors or conditions 8
1.4.5 Economic Burden 8

2 EVIDENCE FOR CVD PREVENTION 9

High-risk approach 9
Population-wide approach 9
Screening and stratification 10
Secondary prevention 10
Whole system integrated approach to CVD risk factor prevention 10
Evidence for self-management 11

3 SITUATION AND GAP ANALYSIS 12

3.1 Surveillance and monitoring 12

3.1.1 Current Status 12

3.2 Primary prevention: health promotion and risk awareness 13

3.2.1 Current Status 13
3.2.2 Gaps 13

3.3 Secondary prevention: screening and aligned healthcare response across primary, secondary and tertiary care 13

3.3.1 Current Status 13
3.3.2 Gaps 14

3.4 Whole-system approach to CVD prevention 14

3.4.1 Current Status 14
3.4.2 Gaps 14

4 SELF ASSESMENT 15

5 PREVENTING CARDIOVASCULAR DISEASES IN QATAR 2019-2022 19

5.1 Vision 19
5.2 Goals and Outcome 19
5.3 Overall Strategy Milestones: 19
5.4 SIX STRATEGIC PILLARS OF NATIONAL CVD PREVENTION STRATEGY 20
Strategic Pillar 1: Awareness and Prevention 20
Awareness 21
Prevention 22
Strategic Pillar 2: Patient empowerment 24
Strategic Pillar 3: Care delivery 26

Strategic Pillar 4: Human capability and capacity building 30
Strategic Pillar 5: Information Management 31
Strategic Pillar 6: Research 32
5.5 IMPLEMENTATION PLAN 34
5.5.1 Governance 34
5.5.2 Accountability, Monitoring & Reporting 35
5.5.3 Communication and Engagement 36
5.5.4 Key deliverables & Timeline 36
5.5.5 Pilot to scale framework 37
6 BIBLIOGRAPHY 38
7 APPENDICES 40
7.1 Appendix A 41
7.2 Appendix B 43
7.3 Appendix C 45
7.4 Appendix D 46
7.5 Appendix E 47
7.6 Appendix F 49
8 GLOSSARY 52

1 SCOPE

The scope of this strategy is described in the sections below:

1.1 Aim, Rationale and definitions

The aim of this public health strategy is to reduce premature death and disability from Cardiovascular Diseases (CVD) amongst the population of Qatar, through promotion of healthy habits, early identification of those at increase risk of CVD and reduction or reversal of this risk.

The rationale behind this strategy is to provide a roadmap for prevention of cardiovascular disease and risk factors over the next few years. A comprehensive strategy will help align and priortize various CVD-related initiatives within the Ministry of Public Health and across various stakeholder agencies and providers to achieve maximum impact in reduction of CVD related ill-health.

The World Health Organization (WHO) defines CVDs as a broad group of disorders of the blood vessels i.e. coronary heart disease (e.g. angina, myocardial infraction), cerebrovascular disease e.g. transient ischemic attack, stroke) and peripheral vascular diseases. In majority of these cases, the underlying pathology is atherosclerosis, i.e. deposition of fat, in the walls of blood vessels. Atherosclerosis starts to develop in the human body over many years, often beginning in early childhood. It is usually advanced by the time symptoms occur in middle age (35-55 years). Abnormal heart rhythm disorders such as atrial fibrillation also increase the risk of CVD but are rarer. They are not covered in this strategy as they have been addressed in the clinical practice guidelines.

1.2 Risk Factors

The debilitating and often fatal complications of CVD are usually seen in adults and the elderly. However, atherosclerosis – the main pathological process leading to CVDs - begins early in life and progresses gradually through adolescence and early adulthood. It is usually asymptomatic for a long period of time. The rate of progression and severity of atherosclerosis is influenced by very common yet modifiable risk factors such as smoking, unhealthy eating, lack of exercise etc (Figure 1). Continuing exposure to these risk factors leads to further progression of atherosclerosis, resulting in narrowing of blood vessels and breaking away of unstable atherosclerotic pieces (plaques) which can obstruct blood flow to vital organs, such as the heart and the brain leading to heart attack (acute myocardial infarction) and stroke.

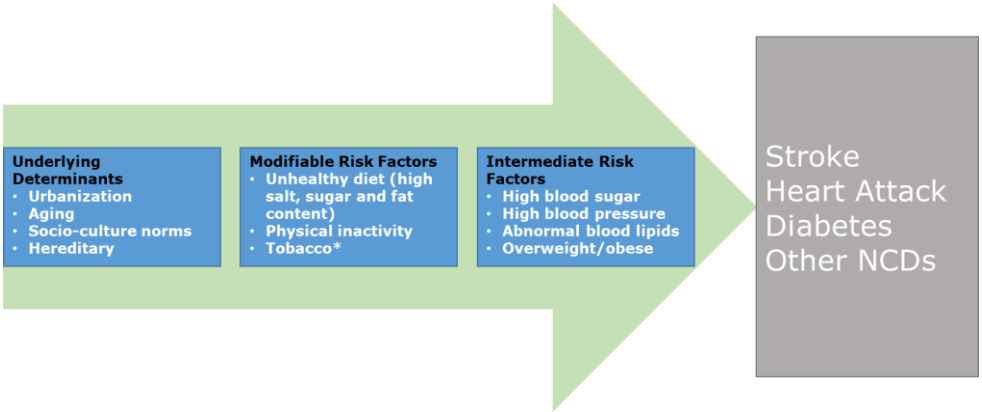


Figure 1: Common factors of cardiovascular disease and other chronic diseases

One of the most important advances in CVD research has been the identification of risk factors associated with the disease. The INTERHEART study examined more than 27,000 people from 52 countries and found that more than 90% of the population-attributable risk for heart attack (acute myocardial infarction, AMI) can be explained by nine potentially modifiable risk factors: ratio of fat-carrying proteins in blood (apolipoprotein B/apolipoprotein A ratio), abdominal obesity, smoking, fruit/vegetable consumption, physical activity diabetes, hypertension, psychosocial factors, and alcohol consumption[1]. Further studies have shown that modification of these risk factors significantly improves cardiovascular health.

1.3 Strategic Context

Cardiovascular diseases are a common global issue especially in the affluent nations. According to the World Health Organization (WHO), CVD are one of the leading cause of death worldwide, accounting for 17.9 million premature deaths every year in people under 70 years of age[2]. This is projected to increase to more than 23 million by 2030[3], with the largest increase to occur throughout the Middle East, due in part to the adoption of the sedentary Western lifestyle. In 2011, the United Nations (UN) recognized this challenge and had set a target to reduce the risk of premature NCDs death by 25% by 2025. UN Sustainability Development Goal 3 has set a revised target for member nations to reduce premature mortality from NCDs by one third[2].

Qatar is quickly becoming one of the leading health systems in the world with a network of primary healthcare services and a world-class acute care sector with multi-speciality hospitals and highly accessible urgent & emergency care services.

However, lifestyle related chronic diseases like diabetes, hypertension, stroke and myocardial infraction are on the rise. The National Health Strategy 2018-2022 notes that the incidence of non-communicable diseases (NCDs) within the local population is growing, largely due to unhealthy lifestyle habits such as physical inactivity, unhealthy diet and tobacco usage[4].

A review of heart attack and stroke patients admitted to the Hamad Medical Corporation between June 2006 and June 2008 found that more than two-thirds of heart attack and half of stroke cases occurred in people under 55 years of age with many cases occurring in people as young as 40 years of age. These cases were predominantly males and Qatari nationals. Diabetes was found to be the strongest underlying risk factor (with diabetics at 3 to 4 times increased risk of CVD compared to those without diabetes), followed by high blood pressure (approximately 2 to 3 times increase in risk). Approximately 40% of patients were overweight and an additional 30% were obese. Lack of regular physical activity and smoking were also shown to significantly impact[5].

The Ministry of Public Health in Qatar launched the National Health Strategy (2018-2022) (NHS II), a comprehensive suite of reforms aligned to the Qatar National Vision (QNV) 2030 that will advance Qatar's Healthcare Vision of creating a world-class, patient-centered healthcare system (Figure 2). Aligned to NHS II is the Qatar Public Health Strategy (PHS) (2017-2022) which aims to address cardiovascular and diabetes health areas among a total of 16 key health areas.

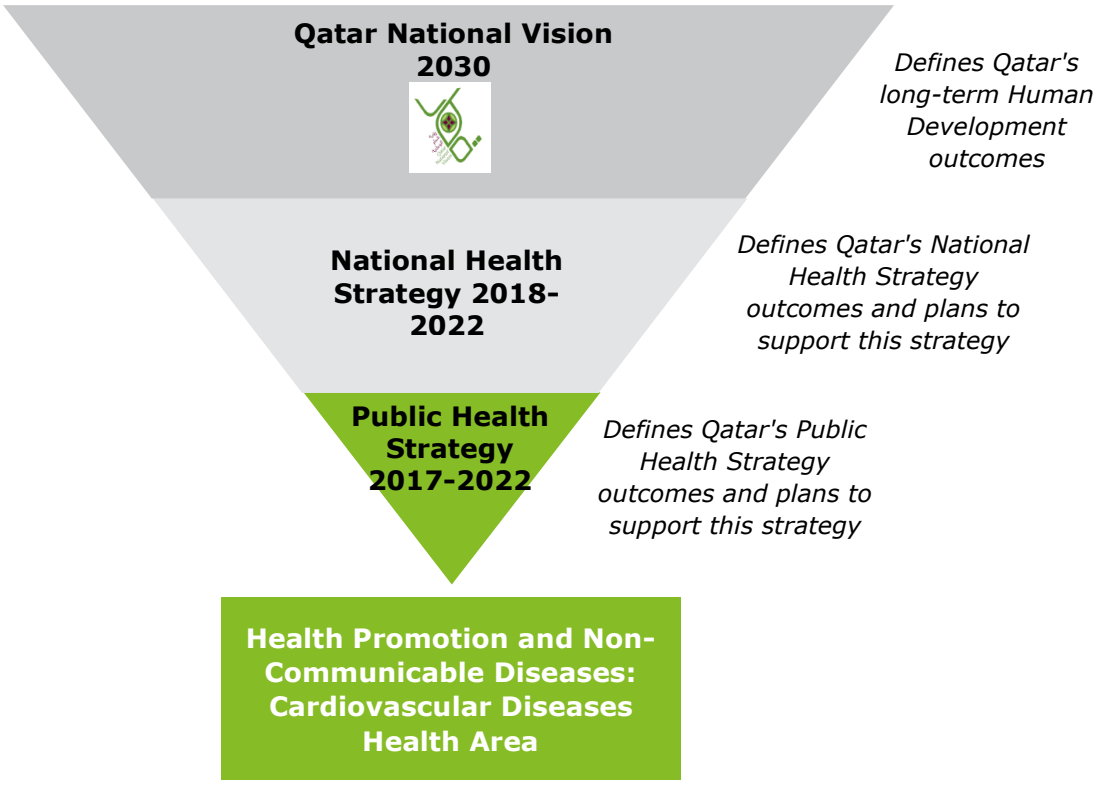


Figure 2: Qatar Nation Vision 2030 – path for a prosperous and healthy Qatar

Prevention and control of NCDs is on the national health agenda for Qatar. This is reflected in the strategic initiatives and targets set out in the National Health Strategy, summarised below:

Figure 3: National Health Strategy (2018-2022) – Enhanced health promotion and disease prevention

The Qatar Public Health Strategy 2017-2022 has strengthened the efforts to promote population-wide primary prevention of non-communicable diseases, through the implementation of the diabetes strategy, tobacco control measures and campaigns targeting diet and physical activity. These activities target common risk factors that are shared by CVD, cancer, diabetes and chronic respiratory disease, and their implementation is critical if the growing burden of these non-communicable diseases is to be controlled.

The Cardiovacular Disease Prevention Strategy will lead the delivery of this national vision and commitment to reducing CVD burden. Magnitude of risk factors in the resident population and the strong evidence in favour of public health measures aimed at primary prevention, early diagnosis and timely treatment, indicates that successful implementation of this CVD strategy will help to significantly reduce burden of CVD and improving overall health of the population.

1.4 Burden of Disease

1.4.1 Mortality and premature mortality

Cardiovascular deaths in Qatar are as high as in other high-income countries. In 2012, age adjusted mortality in Qatar was 151.2 and 159.7 per 100,000, respectively for men and women. In the US in the same period, CVD mortality was 193.2 per 100,000[6]. The proportion of deaths due to CVD is also comparable although slightly lower than the proportion of deaths in the UAE, where CVD accounts for 30% of all deaths[7]

CVDs account for the second largest cause of premature mortality in Qatar as seen in the below analysis from the WHO Global Burden of Disease Study in 2010.

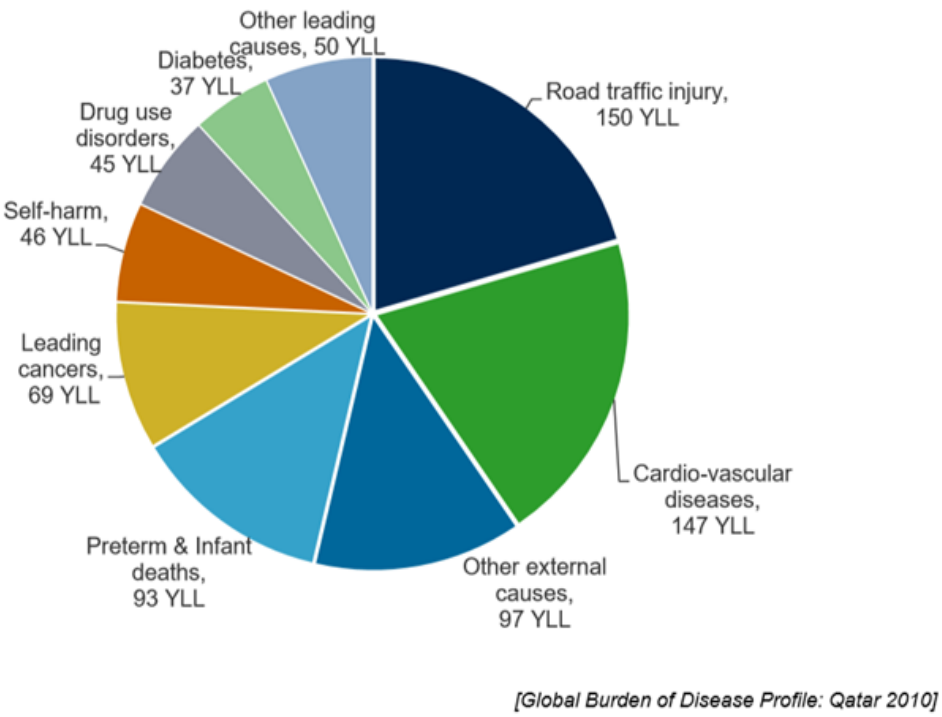


Figure 4: Years of life lost of leading causes of death responsible for 70% of premature mortality

Risk of CVD and CVD-associated death increases with age. During 2011-2013, out for every 100,000 population nearly eight Qatari males and four non-Qatari males aged between 20-44 years, died of diseases of the circulatory system each year. After age 45, CVD death rate rose to 247 per 100,000 per year for Qatari men[7]. While women have relatively lower risk than men, the same pattern of increase in risk with age was seen for them. In 2011-2013 review, among women, CVD death rates rose from nearly 2 (2.1-1.6) per 100000 (for Qatari and non-Qatari women) 20-44 years to nearly 150 deaths per 100000 for Qatari women after 45 years of age.[7]

1.4.2 Prevalence

Data from the first comprehensive national NCD risk factor survey (STEPwise survey 2012) indicates that approximately 3 out of every 4 of adult Qataris aged 18-64 years have one to three CVD risk factors and slightly more than half of the survey respondents had three to five risk factors each[8]. The most common risk factors are poor dietary habits, physical inactivity, smoking, obesity, high blood pressure (hypertension) and high body fat (dyslipidemia), as presented in Figure 5.

Additionally, the respondents aged 45 and above are at higher risk of having these risk factors compared to those in the age group 18 to 44 (70.4% vs 44.9%, respectively). The prevalence of obesity (body mass index [BMI] ≥30 kg/m^2) in the adult population is estimated to be 44.2%. Nearly 33% of the adult population has elevated blood pressure (systolic 140 mmHg and/or diastolic 90 mmHg) or are currently on medication for raised blood pressure.[8]

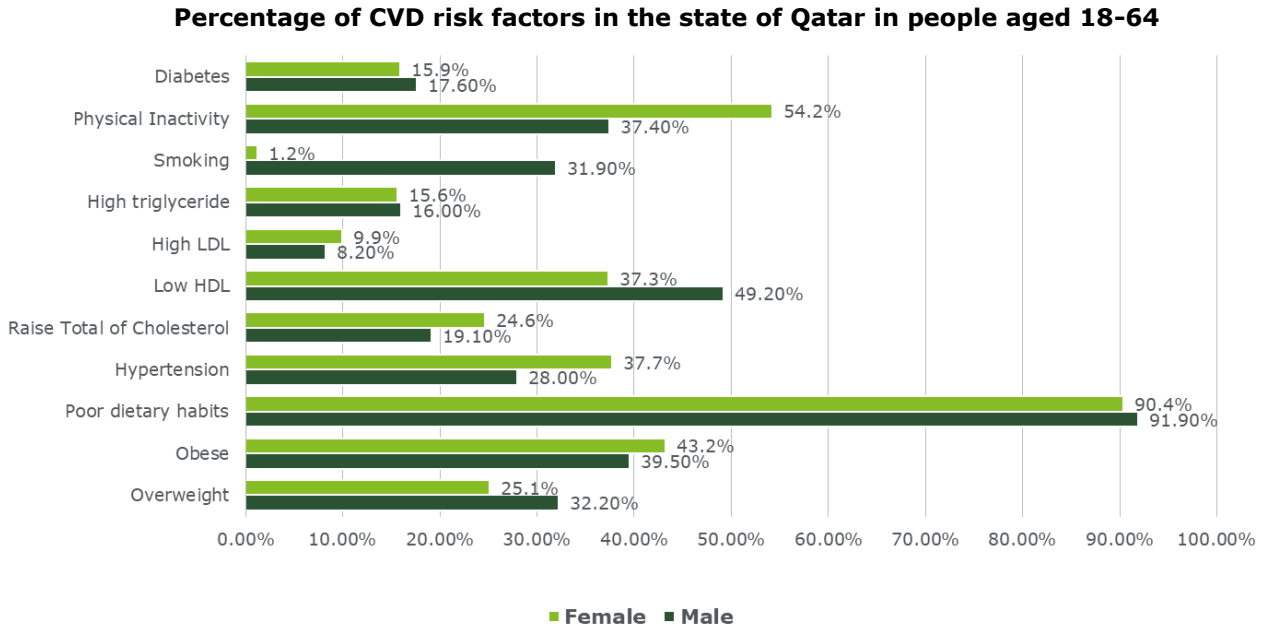


Figure 5: Relevance of key modifiable risk factors distribution

1.4.3 Demographic factors

There is an age and gender difference in risk factor profile with men and people in older age groups at higher risk of having underlying risk factors and developing CVD. However, the roots of CVD risk and unhealthy lifestyle habits are laid early in childhood. For e.g. studies have reported that obesity prevalence in children and young adults (<30 years old) is higher than expected for the Qatari population and increasing (from 37% to 47% for Qatari boys and from 24% to 41% for Qatari girls in six years from 2003 to 2009)[9]. It has also been reported that a large proportion of the population are developing risks for CVD before their 30th birthday. More than half (57%) of obese and overweight Qataris are less than 30 years of age[9,10]. Qatar has a growing and disproportionately large subpopulation of younger people. With 66% of the Qataris being less than <30 years of age[11], it is important to focus CVD prevention initiatives on this subgroup, along with the adults.

1.4.4 Patients with undiagnosed risk factors or conditions

A significant proportion of Qatari population have undiagnosed risk factors. STEPwise Survey 2012 reported that a significant proportion of adults aged 18-64 years had undiagnosed risk factors or intermediary conditions that were not managed to prevent further progression of CVD. E.g. 13.4% of previously undiagnosed respondents were found to have raised blood pressure (SBP ≥140 and/or DBP ≥ 90 mmHg) during the survey[8]. Amongst those with known hypertension, only 66.6% were receiving treatment for high blood pressure prescribed by a doctor and 32.9% of had high blood pressure when measured during the survey. These rates were dramatically higher after the age of 45 years and amongst men. Similarly, for diabetes, only 62 % of the diabetic respondents reported conducting regular visits to a healthcare facility for diabetes[8].

1.4.5 Economic Burden

Cardiovascular diseases rank the most costly diseases globally[12]. According to a recent study, the total direct and indirect costs calculated for the five selected NCDs in the GCC amounted to approximately 36.2 billion US dollars in 2013. It represents 1.5 times the reported direct health care spending in the region[13].The same study estimated, that without public health and policy interventions, the latter expense is expected to rise up to 68 billion US dollars per year by 2022 for the same region. It also predicted that the total cost per head for NCDs is higher in Qatar, estimated to be approximately $2,778 per person compared to $603 per person in Oman.

2 EVIDENCE FOR CVD PREVENTION

According to the World Health Organisation, at least one-third of the world population is currently at high risk of developing major cardiovascular events: an estimated 1.3 billion due to tobacco use, 1 billion due to overweight and at least another billion due to elevated blood pressure, blood cholesterol and/or diabetes. A modest reduction in these risk factors, achieved through an appropriate combination of population-wide (primary prevention) and high-risk strategies (secondary prevention), has the potential to halve cardiovascular events within a relatively short period of five years.[14]

An extensive modelling of Global Burden of Disease study data has shown that decreasing the prevalence of hypertension accounted for the largest risk reduction, followed by a reduction in tobacco smoking for men and obesity for women.[15]

Evidence from multiple large cohort studies[1,16,17] has shown success with a 2-pronged approach to public health action for CVD prevention as follows:

High-risk approach

Early diagnosis of CVD risk factors, absolute risk stratification and active interventions targeted at individuals most likely to benefit from individualised prevention and treatment plans.

Population-wide approach

Aimed at generally supporting healthy lifestyle changes at a population level through a mix of public health, educational, social and environmental interventions[14]. Although cardiovascular associated morbidity and mortality are less likely to occur in low risk people, there is not a risk level considered as "safe". Without public health interventions targeting the population as a whole, CVD events will occur in the majority of people, mainly people with low and moderate risk levels. Public health efforts can successfully slow down the development of atherosclerosis (simultaneously reducing the incidence of associated cancers and chronic respiratory disease) in young people. This will reduce the likelihood of future CVD events and epidemics, such as the ones see in 1960 to 1990 in many high-income countries. Population wide strategies will also aim at addressing other risk factors. For example, they will target lifestyle modification in high risk people, leading to health promoting behaviours.[14]

While high-risk strategies will not prevent cardiovascular events in individuals with modest elevations of several risk factors, population-based strategies by themselves will not be able to prevent cardiovascular events in those at high risk. The two approaches are complementary and synergistic.

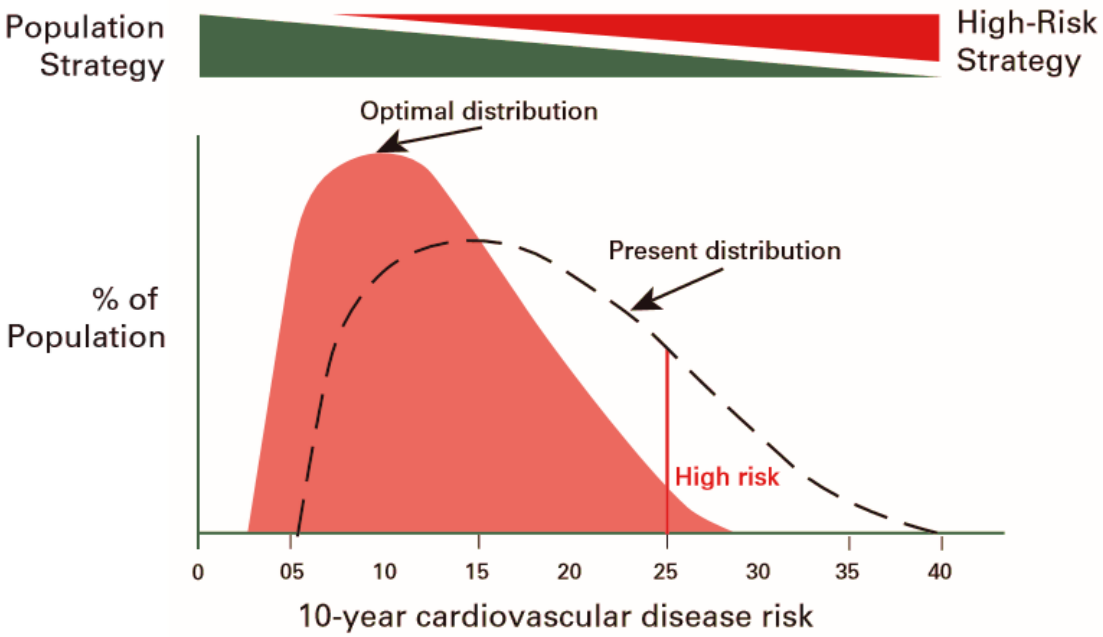


Figure 6: Cardiovascular disease prevention. Translating evidence into action. Geneva, WHO, 2005.

Screening and stratification

Screening and risk stratification have been proven in large epidemiological and clinical studies as the best way to optimise interventions at individual level based on overall risk. This total risk approach is based on the evidence for continuous relationship between CVD risk factors and cardiovascular diseases; and that risk factors tend to appear as multiple co-morbidities in the same population groups. Use of risk prediction charts to estimate total cardiovascular risk is a major advance on the older practice of identifying and treating individual risk factors, such as hypertension and hypercholesterolemia. This approach has now been widely taken up in cardiovascular prevention guidelines in most high-income countries including US, UK, most Nordic countries, Australia and New Zealand[17,18,19,20]. In the absence of locally validated risk-prediction tool, the current guidelines in Qatar recommend the American College of Cardiology/American Heart Association (ACC/AHA) Pooled Cohort Equations to be used to estimate 10-year ASCVD risk in appropriate individuals with and without diabetes. The evidence base for ACC/AHA equation is derived from various large cohort studies, most commonly the Framingham Heart Study[21]. These equations have been widely tested in North American and European populations of European origin and validated only in one Asian study[22]. A recent Korean study found that for the modified KRPM score showed better predictive value for ASCVD risk prediction in a Korean population than the ACC/AHA Equations applied directly.[23]

Secondary prevention

To complete the prevention cycle, secondary prevention, aimed at people who have survived an acute CVD event is equally important. People with history of a cardiovascular event need the highest level of active behavioural (dietary, lifestyle) and clinical (pharmaceutical and /or surgical) interventions to manage the underlying risk factors. The risk of recurrence of a CVD event in survivors is significantly high and studies have shown significant reduction in this risk with active control of blood sugar levels, lowering of blood pressure, weight loss and smoking cessation[24].

Whole system integrated approach to CVD risk factor prevention

Addressing lifestyle change and social determinants of health is key to primary prevention of CVD. An environment that is conducive to making healthy choices affects the likelihood that people will achieve positive behaviour change, compliance with lifestyle and medical interventions[25,26]. In addition, schools, workplaces, religious and social institutions have a place in providing the right culture and support for sustaining change.

An effective CVD prevention strategy, while driven by public health department, needs to be based on a whole–system approach with shared accountability across the stakeholders.

A comparative effectiveness review of public health services in England[27] noted that preventative investment for health improvement has focused too much on the 'medical' aspects of disease prevention, detection and management, and not enough on the social, cultural and economic factors that cause the disease 'risk conditions' in the first place and are instrumental to sustaining lifestyle changes.

Figure 7: Multi-pronged approach and multi-stakeholder approach to cardiovascular disease prevention, adapted from the Qatar Public Health Strategy 2017-2022

Evidence for self-management

Self-management support interventions such as the Stanford Chronic Disease Self-Management Program (CDSMP) are becoming more widespread to help individuals self-manage chronic diseases and adopt healthy behaviours. A pooled metanalysis of randomized controlled trials comparing self-management support interventions for general chronic disease against usual care found that the CDSMP led to statistically significant, short-term improvements across a number of health status measures (including some measures of health-related quality of life), healthy behaviours, and self-efficacy compared to usual care. There was not enough evidence to show reduction in healthcare utilization[28]. When implemented in Singapore and China (Shanghai) the CDSMP was acceptable culturally to the patients. The programme improved participants health habits, self-mangement and reduced healthcare services utilisation, especially acute care[29,30].

3 SITUATION AND GAP ANALYSIS

3.1 Surveillance and monitoring

3.1.1 Current Status

Non-communicable diseases prevalence baseline in Qatar was established in 2006 with the World Health Survey (WHS). This was followed by the more comprehensive Qatar STEPwise Survey 2012, which would be repeated in 2020. In addition, some specific CVD related data is available from Global Adult Tobacco Survey (GATS) and topical health surveys by academic researchers. Other key sources of CVD–relevant data is available from registration of deaths, comprehensive height and weight information available from school health records.

A recent mapping of the core strategic initiatives under the current National Public Health Strategy and the key planned projects in each of the Health Areas showed that a significant number of surveys and surveillance initiatives are planned over the next 2-3 years. The latter are aimed at measuring varying aspects of common NCD risk factor baselines, changes or measuring impact of interventions such as increase in knowledge, attitude and practice. A sample of this mapping is included in the Annex A.

Limited risk factor data

The STEPwise surveys are gold standard for providing representative and comprehensive population-based data on cardiovascular and other key NCD risk factors, including physical and biochemical measures. However, such extensive surveys are time consuming and resource intensive. Hence, Qatar is currently in the midst of doing its second STEPwise survey, seven years after the first one.

Currently, there is a lack of systematic interim facilities-based surveillance systems (e.g. hospital and/or primary-care based disease registers) or population-based self-reported surveillance systems (e.g. US Behavioural Risk Factor Surveillance System, BRFSS). These systems can help identify interim changes to the risk factor trends, detailed information on specific aspects of CVD management and impact of public health interventions.

Lack of health information linkage

Despite majority of healthcare information in primary and acute care being available in digital form in Qatar and largely on the same digital platform, there is very little data linkage across the health sector due to technical and inter-sectoral challenges.

Gaps in planning for information

There is a need to plan for and build monitoring and evaluation components in both public health programmes and across the healthcare system. Currently most public health interventions for e.g. increasing average daily physical activity or reducing dietary salt and fat intake do not include a pre and post-implementation evaluation. Similarly, there are gaps in planning for health information that will be useful not only for individual patient management but also for population health purposes in the primary care and hospital setting.

Limited data driven intelligence for public health decisions

Despite the limitations discussed above, there is information available in public policy space, which can benefit from deeper epidemiological dives to support effective CVD policies and interventions. However, this has currently been limited by technical challenges in data pooling, cleaning and analysis due to limited epidemiological and statistical expertise.

Information integration

Gap in linking epidemiological information to drive CVD research in areas of public health interest and in a local population context, such as:

- Large genotypic and phenotypic cohort studies to identify genetic and environmental impact on risk factors,
- Key drivers of burden of cardiovascular disease, and
- Validation of risk factor prediction tools.

3.2 Primary prevention: health promotion and risk awareness

3.2.1 Current Status

The risk factors for CVD are common across the major NCDs including cancer and chronic respiratory disease as well as intermediary risk factors (Table 1). While the Qatar National Diabetes Strategy 2016-2022 was the first national policy to develop a comprehensive and systematic approach for primary prevention and early detection of common NCD risk factors, there has been a concerted public health effort led by the Ministry of Public Health (previously the Supreme Council for Health) to address the rising epidemic of non-communicable Diseases in Qatar over the last decade. Hence, there are multiple programmes in place to address either a specific disease such as diabetes, hypertension and stroke or risk factor prevention such as anti-smoking and healthy lifestyle. These programmes comprise of awareness and education campaigns, regulatory measures such as anti-tobacco legislation and those targeted at industry, such as food labelling.

The summary of key initiatives in the current national public health strategy that directly or indirectly impact CVD prevention are enclosed in Annex B.

3.2.2 Gaps

Limitation of approach and scope

The public health programmes are largely based on general population-based approach without screening and targeted programmes for high-risk population.

Limited alignment across programmes

The public health programmes for CVD risk factors and other NCDs have traditionally been vertically organised around individual risks or diseases. This has led to difficulty in developing a consistent and sustainable public awareness and engagement strategy targeted at common modifiable NCD risks.

Communication challenges

Awareness campaigns, though in local languages, are not always tailored to the various population segments and specific socio-cultural aspects of health risk behaviours. Communication largely through formal governmental platforms and this do not always reach and engage the varied population groups.

Limited community and public partnership

Health education is largely didactic with limited active community engagement.

Gaps in evaluation

Effectiveness of various campaigns is measured largely in terms of activity output indicators such as attendance and event feedback.

3.3 Secondary prevention: screening and aligned healthcare response across primary, secondary and tertiary care

3.3.1 Current Status

Besides MoPH, the primary healthcare sector in Qatar, largely served by the PHCC, and the acute care sector include the Hamad Medical Corporation (HMC) and Sidra are key stakeholders in the full-spectrum care for Cardiovascular Diseases. These organisations work together at a policy level through national committees

such as National Diabetes Committee. The terms of reference and membership of the National Diabetes Committee are included in Annex C.

There are clinical practice guidelines for various aspects of clinical management of CVD including individual risk factors and the entire spectrum of cardiovascular diseases. A list of the current CVD-related guidelines is enclosed in Annex D.

3.3.2 Gaps

Lack of organised health screening for CVD risk factors

The current screening in primary care setting for cardiovascular risk factors (diabetes, hypertension, and lipid disorders) is on opportunistic or symptomatic basis.

Active screening

Despite strong evidence that the risks tend to run in family and social clusters due to common lifestyle practices and in some cases genetic predisposition such as familial hyperlipidemia, there is a lack of case-finding approach for active screening of families of patients diagnosed with CVD risk factors. This is a missed opportunity for early detection and prevention of progression to subclinical and clinical disease.

Inconsistent approach to health advocacy and referral to behavioural support services

Advocacy and referral for behaviorial risk factor modification such as diet for especially in primary but also in acute and post-acute settings is inconsistent.

Lack of clarity on adoption of clinical practice guidelines (CPGs)

In the absence of CPG audits, mandated compliance with the CPGs and no requirement for the providers to make changes to service delivery and care pathways in line with practice guidelines, the level of standardisation in clinical care especially secondary prevention is not known.

Gap in total care-continuum approach to CVD management

Patient follow-up and referrals across the spectrum of care are not well linked. The information on risk factors and outcomes of services such as smoking cessation are not shared well between PHCC and acute care. Similarly, continuity of step-down care is a problem. Patients with CVD often appear in acute sector as an emergency admission for stroke or cardiac arrest. Once treated and discharged, there is inconsistency in prevention of future CVD events. These patients require intensive secondary and tertiary prevention that includes drastic lifestyle modification in addition to therapeutic interventions, when discharged back to the primary care and community. This hand-over of care and enabling self-management by patients and their families, is currently inconsistent.

Limited patient empowerment:

Limited focus on patient empowerment with focus largely on medically led and delivered interventions.

3.4 Whole-system approach to CVD prevention

3.4.1 Current Status

There is engagement at the inter-ministerial level for most major NCD areas, with key community/religious, education, trade and cultural stakeholders represented on the respective taskforces and workgroups. However, Public Health department is currently driving and implementing most policies.

3.4.2 Gaps

Socio-cultural and economic determinants of CVDs are well established. While people can be made aware of changes needed to their lifestyle through public health messaging, a coordinated nationwide and multi-agency approach is needed to help them adopt and sustain these changes[31].

4 SELF ASSESMENT

The WHO CHOICE analysis assessed and categorized 88 interventions (published in peer reviewed journal with demonstrated and quantifiable effect size) based on their feasibility and cost-effectiveness ratio (expressed as International dollars (I$) per disability adjusted life year (DALY)) of ≤ I$ 100 per DALY averted in low- and lower middle-income countries (LMICs); cost-effectiveness ratio > I$ 100 per DALY averted; and those for which WHO CHOICE analysis could not be conducted.

The absence of WHO-CHOICE analysis does not necessarily mean that an intervention is not cost-effective, affordable or feasible – rather, there were methodological or capacity reasons for which the WHO-CHOICE analysis could not be completed at the current time. The interventions are categorised as follows:

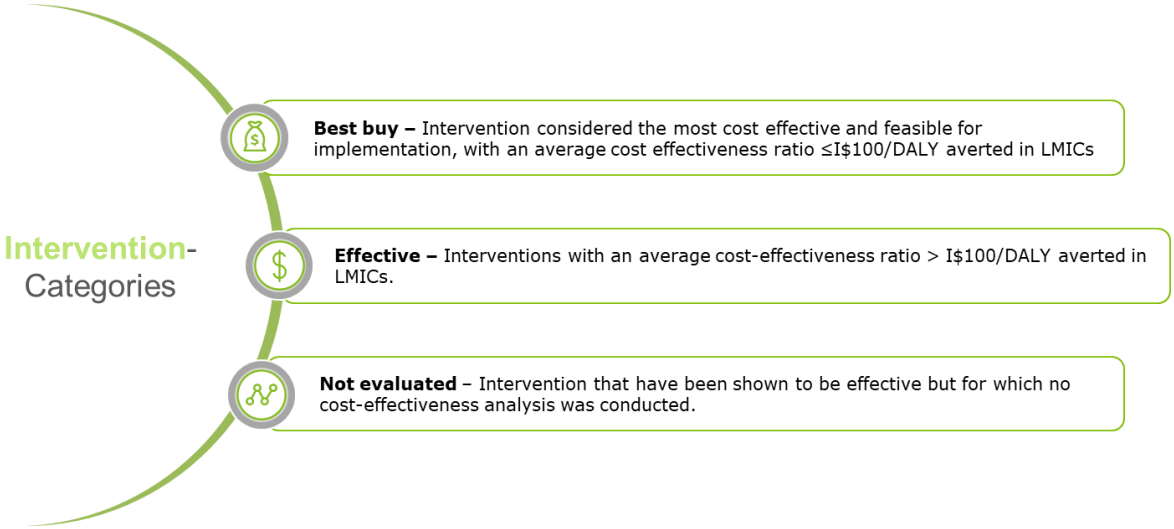


Figure 8: Intervention categorization, 'BEST BUYS' AND OTHER RECOMMENDED INTERVENTIONS FOR THE PREVENTION AND CONTROL OF NONCOMMUNICABLE DISEASES, WHO, 2017

A self-assessment of current public health interventions in Qatar vis-a-vis WHO recommended 'best buys' and other recommended interventions, aimed at cardiovascular and other NCDs common risk factor prevention.

Table 1a: Unhealthy diet

	WHO recommended interventions	***Local situation***
Best buys	*Reduce salt intake through the reformation of food products*	
	Reduce salt intake through the establishment of a supportive environment in public institution such as hospitals, schools, workplaces and nursing homes.	
	Reduce salt intake through a behaviour change communication and mass media campaigns	
	Reduce salt intake through the implementation of front-of-pack labelling	
Effective Interventions	*Eliminate industrial trans fat through the development of legislation to ban their use in the food chain*	
	Reduce sugar consumption through effective taxation on sugar-sweetened beverages. (already in place)	
Other recommended interventions	*Promote and support exclusive breastfeeding for the first 6 months, including promotion of breastfeeding*	
	Implement subsidies to increase the intake of fruit and vegetables	
	Replace trans fats and saturated fans with unsaturated fats through reformulation, labelling, fiscal policies	
	Limit portion and packages sizes	
	Implement nutrition education and counselling in different settings (e.g. in preschools, schools workplaces and hospitals) to increase the intake of fruit and vegetables	
	Implement nutrition labelling to reduce total energy intake, sugar, sodium and fats	
	Implement mass media campaigns on health diet to reduce the intake of total fat, saturated fats, sugar and salts, and promote healthy diets	

Label	
	Policies in place and implemented
	Pending implementation
	Not covered under current policies

Table 1b: Physical inactivity

	WHO recommended interventions	Local situation
Best buys	Implement community wide public education and awareness campaigns for physical activity	
Effective Interventions	Provide physical counselling and referral as part of routine primary healthcare services	
Other recommended interventions	Implement whole-of-school programme that includes quality physical education, availability of adequate facilities and programmes	
	Ensure that macro-level urban design supports active transport strategies	
	Provide convenient and safe access to quality public open space and adequate infrastructure to support walking and cycling	
	Implement multi-component workplace physical activity programmes	
	Promotion of physical activity through organised sport groups and clubs, programmes and events	

Label	
	Policies in place and implemented
	Pending implementation
	Not covered under current policies

Table 1c: Tobacco Control

	WHO recommended interventions	Local situation
Best buys	Increase excise taxes and prices on tobacco products	■
	Implement plain/standardised packaging and/or large graphic health warnings on all tobacco products	■
	Enact and enforce comprehensive bans on tobacco advertising, promotion and sponsorship	■
	Eliminate exposure to second-hand smoke in all indoor workplaces, public places and public transport	■
	Implement effective mass media campaigns that educate the public about the harm of smoking/tobacco use and second-hand smoke	■
Effective Interventions	Provide cost-covered, effective and population-wide support (including brief advice, toll-free quit line service) for tobacco cessation services to all those who want to quit	■
Other recommended interventions	Implement measures to minimise illicit trade in tobacco products	■
	Ban cross-border advertising, including modern means of communication	■
	Provide mobile phase based tobacco cessation services	■

Label	
■	Policies in place and implemented
■	Pending implementation
■	Not covered under current policies

5 PREVENTING CARDIOVASCULAR DISEASES IN QATAR 2019-2022

The national cardiovascular disease strategy lays down the overall vision, goals and strategic initiatives for the period of 2019-2022 aimed at overcoming the gaps in cardiovascular disease and risk factor prevention in Qatar.

5.1 Vision

To reduce premature death and disability from Cardiovascular Diseases in Qatar by widespread adoption of healthy lifestyle practices, early identification and timely treatment of those at higher risk of developing these diseases, though a coordinated whole-system approach across health and the broader social sector, patients, general public and communities.

5.2 Goals and Outcome

The National Health Strategy 2018-22 has set a target for 15% reduction in premature mortality from major NCDs by 2022. In support of this target, through the implementation of this strategy, we will decrease the incidence of diabetes, hypertension, and dyslipidaemia and reduce premature mortality from cardiovascular complications of these diseases such as stroke and heart attacks, resulting in better health and quality of life.

Qatar's National Cardiovascular Disease Strategy is built on the following five areas:

1. Increase public awareness and improve patient empowerment to adopt healthy lifestyle practices to prevent development of CVD risk factors i.e. diabetes, hypertension and dyslipidaemia.
2. Early diagnosis of cardiovascular risk factors to reduce the occurrence of stroke and heart attack, and premature mortality from these diseases.
3. Deliver high quality and coordinated cardiovascular care across primary, secondary and tertiary health system.
4. Strengthen cardiovascular surveillance through use of information technology and analytics.
5. Develop public health research capacity and capability for investigation into cardiovascular diseases and risk factors.

5.3 Overall Strategy Milestones:

By 2020: full implementation of the national cardiovascular disease-screening program.

By 2020: All target adult populations will be screened for diabetes, hypertension and dyslipidaemia

By 2020: all screened at-risk population and patients will be offered an annual health check ups

By 2021: all healthcare professionals will receive continuing education in cardiovascular risk factor prevention, management and care-pathways

Other milestones for consideration by NCD division:

By 2020: New thematic national healthy lifestyle campaign (to replace various NCD campaigns) with all relevant stakeholders involved.

By 2019: Public guidelines on cardiovascular disease prevention, screening and management.

By 2020: Implement

the public Health thematic research grant programme.

By 2021: launch of the national cardiovascular disease management programme (CDMP), which provide a comprehensive, evidence-based care continuum package for CVD prevention and management to all at-risk patients on annual health plan.

By 2022: launch of the national cardiovascular diseases management database (CDMD) which will track patient and practice level outcomes for quality of care (HbA1C, annual diabetic foot and eye screening, LDL, HDL, cholesterol, blood pressure, weight, etc.)

By 2022, Achieve 70% recruitment and training of the required healthcare professionals, community champions, and lifestyle and healthcare counsellors to meet the needs of the high-risk and patient populations.

5.4 SIX STRATEGIC PILLARS OF NATIONAL CVD PREVENTION STRATEGY

Qatar National Cardiovascular Disease Prevention Strategy is designed to achieve the future vision to prevent cardiovascular risk factors and this reduce death and suffering from heart disease, stroke and diabetes in the State of Qatar. We will achieve this through six strategic pillars which will address the current gaps in the CVD prevention.

Pillar 1: Awareness and Prevention

Pillar 2: Patient Empowerment

Pillar 3: Care delivery

Pillar 4: Human Capability and Capacity Building

Pillar 5: Information Management

Pillar 6: Research

Strategic Pillar 1: Awareness and Prevention

This is the primary prevention strategic pillar for strengthening the public health actions aimed at reducing the risk of developing cardiovascular diseases in general population.

The aim of this strategic pillar is to provide a co-ordinated approach to the multi-pronged primary prevention initiatives directly and indirectly linked to cardiovascular disease prevention, across the MoPH. This includes health education campaigns for diabetes, heart disease, stroke, healthy lifestyle etc.; specific prevention programs such as smoking cessation, health check-up camps, school health screening etc.,; legislation against smoking in public places as well as industry partnerships to provide healthier food options, reducing pollution; and community partnerships to co-develop and implement tailored health programs.

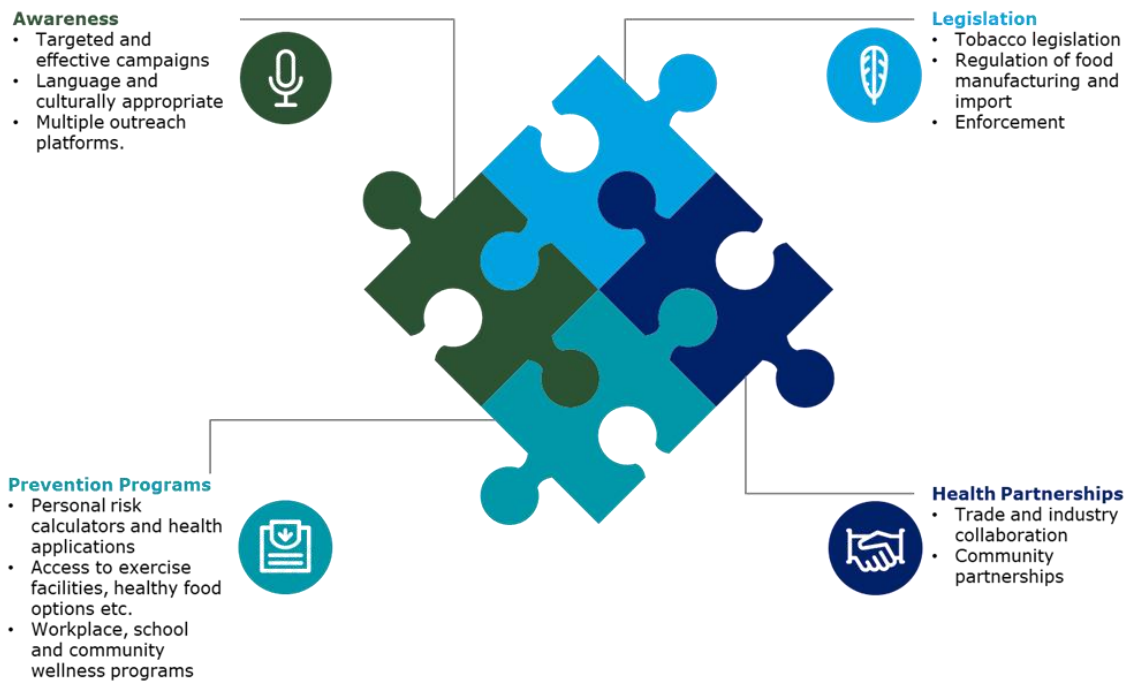


Figure 9: *multi-prolonged strategy for primary & secondary prevention*

Awareness

Effective public awareness programs are key elements in preventing CVD because significant amount of risk can be reversed or reduced by simple changes in daily lifestyle. Hence, there needs to be widespread understanding of CVD risk factors, disease progression over time and their devastating complications.

There are current awareness programs targeting CVD directly or indirectly. These are provided both by MoPH and healthcare providers including HMC and PHCC. But these progams have so far not been very effective. For example, the National Diabetes Strategy has noted that despite the significant number of awareness initiatives, public surveys conducted by the strategy team (SCH, 2-2015) revealed that 63% of surveyed population had never been screened for diabetes only 14% were aware of Type 1 and 41% were aware of Type 2 diabetes.

The reasons behind lack of effectiveness of these awareness programmes include:

- Multiple messages with conflicting information.

- Campaigns not being tailored to high-risk population segments and specific socio-cultural aspects of health risk behaviours.

- Communication through formal governmental platforms, which do nothave the same outreach as new forms of social media.

- Messages and campaigns are largely didactic with limited active community interaction and engagement.

The strategic pillar on CVD awareness will focus on delivering the following:

1. **Standardised and simplified CVD prevention content** across national agencies and providers to provide

    a. Clear and easy to understand key messages to help people remember what they need to do to avoid or control diseases such as diabetes, hypertension and hyperlipidemia and obesity etc.

b. Alignment across messages for other non- communicable diseases with same risk factors.

c. Consistent logos and branding so that messages are recognizable and easily memorable.

d. Active engagement of stakeholder groups such as youth, women, working adults for co-creation of tailored content and appropriate channels for proliferation of information.

2. **Multipronged communication plan** to improve dissemination of messages through

a. Repeated and consistent messages through multiple channels to ensure that the messages reach everyone and are well understood by all.

b. Use of multiple outreach channels including varied and non-conventional communication channels and community engagement platforms such as non-governmental forums and popular social media apps.

c. Use of subliminal marketing for positive messaging through popular TV series etc

d. Use of influencers such as popular political and religious leaders, media personalities and youth champions to influence positive behavioural change.

3. **Active community engagement for prevention** through

a. A combined thematic annual healthy lifestyle campaigns which would be systematically planned series of community events dedicated to health promotion and awareness instead of multiple unlinked campaigns. The annual themes of campaigns would include diabetes/CVD prevention; health screening; exercise and eat well; healthy blood pressure etc. The campaign will begin at the same time every year, with a launch event such as mass exercise led by a national leader.

b. Capitalisation on key community milestones or celebrations such as getting fit for 2022 World Cup, to motivate change and wider community participation.

4. **Translating awareness into action** through

a. Sustained engagement of lifestyle changes with subscription to validated health applications on digital and mobile platforms. Potential development of health tracker applications with technology partner would be considered in the next phase of the strategy.

b. Work with the stakeholders to develop and implement health promoting schools, communities and workplace programs to enable a supportive environment for lifestyle changes[32,33]. This would include

- CVD and common risk factor awareness and healthy lifestyle strategies educational content
- A start-up toolkit and advisory support services for health promoting school workplace, community services etc.
    - Accreditation and awards to recognise health promoting schools, employers and community volunteers etc.

Prevention

The awareness and its translation into adopting healthy lifestyle practice will help reduce the primary risk for CVD in general population. However, there is a significant proportion of population who already have already developed the diseases or risk factors. For example, data from the

national health survey showed a significant number of adults have already developed one or more intermediate risk factors and a large proportion of them are unaware of these underlying conditions. For instance, the National Diabetes Strategy estimates that about one out of six adults in Qatar have pre-diabetes and one in every 3 to 4 diabetics are not aware of their condition and not seeking treatment.

Cardiovascular diseases are chronic progressive i.e. they develop gradually over time, starting with an intermediate risk phase (such as single condition such as pre-diabetes or a multitude of conditions such as metabolic syndrome) before progressing to full-blown disease and eventually complications. As discussed above in section 2, evidence has shown that there are opportunities for timely preventive action, which can lead to reversal of disease in early stages or progression to complication.
The prevention strategic pillar will deliver the following:

1- **Cardiovascular disease screening programme**

Starting from 2020 as a mass-scale catch-up screening drive to estimate the prevalence of the high-risk population, the CVD screening will eventually be part of routine health screening for adult resident population as well as embedded into routine health checks such as employment and workplace health screening programmes. The key parts of this screening programme will be

- **National baseline screening drive**
- **Integrated routine screening programme**
- **Post-screening follow-up and referral pathways for care continuity**

- **National baseline screening**
  The first round of (baseline) screening began in 2020 and will target all adult population, this identifying and stratify the population risk of CVD i.e.

  - Healthy population: those who do not have apparent physical or physiological presentation of risk factors;
  - At- risk population: those with underlying risk or intermediate risk who are overweight/obese, smokers, pregnant; who have significant family history of these diseases; and those who lead largely sedentary lifestyle and eat unhealthy foods on regular basis.
  - High-risk population: those with pre-diabetes, metabolic syndrome, undiagnosed and therefore untreated diabetes, hypertension, with or without hyperlipidemia.

  It will take approximately 2 years to invite and screen the entire eligible population. Given the large number of undiagnosed diabetes, hypertension and other cardiovascular risks in the population[6], it is estimated that this phase of the programme could identify up to half a million local (Qatari and resident) population who are at moderate to high risk of developing CVD within the next ten years.

- **Integrated routine CVD screening**
  Once the baseline screening is completed, the screening will be part of routine health check-ups in primary care settings and other screening opportunities such as the employment health checks. In addition, opportunistic screening will be offered in other healthcare settings and community-based wellness and awareness events to ensure full coverage of the population

  MoPH will work with stakeholders to explore the feasibility of extending health screening requirements for employment, immigration, school and higher education enrolment, registration for health insurance etc.

- **Post-screening follow-up and referral pathways**
  A key aspect of screening effectiveness is the post screening follow-up of patients appropriate to their needs and through the healthcare services to ensure they receive the right care at the right place, at the right time and from the right healthcare.

A planned post-screening follow-up will be implemented as follows:

- People who are considered high-risk will be referred to wellness programs and prevention counselling in a community setting
- People who have been recently screened and diagnosed as having CVD risk factors such as recently diagnosed diabetes, hypertension, pre-diabetes, smokers etc will be referred for health education and care in a primary care setting
- People with uncontrolled diabetes, hypertension, morbid obesity or those who are at risk of/ or have developed complications will be referred for further management by specialists through primary care physicians

2- **Health plan**

Every person found to be at intermediate to high risk of developing cardiovascular disease at screening, will be offered a health plan.

Besides a program of regular primary care consultations, diagnostics and therapeutics, this plan will include interactive risk-factor intervention program (e.g. e-consultations for nutrition and smoking, mobile apps for activity motivation and tracking) for high-risk population identified systematically through health screening.

The screening follow-up and the health plan will require review of service capacity, standardisation of referral pathways and linking up of care. This is further addressed under the care model and health plan in Strategic Pillar 3: Care Delivery.

3- **Tracking and linkage of data across the care pathway**

Enhanced linking up of electronic information across screening, primary and acute care will allow coordination of care, better outcomes and care experience for patients. In addition, disease specific database or registry that capture relevant data for monitoring the impact of screening and other healthcare interventions at population level.

As the first step under this strategic initiative, MoPH will develop a diabetes registry to compile information and monitor all at-risk people, people who are pre-diabetic and people with diabetes, to ensure they are receiving the right support.

This registry will later on be extended to include other CVD risk factors of public health importance.

✓ **Impact and outcomes of this strategic pillar are expected to be as follows:**

Public:

- Clear, understandable information about cardiovascular diseases of public health importance to Qatar and the ways to reduce the risk is easily accessible.
- Every adult resident understands his or her risk for developing CVD and is directed for appropriate follow-up at the right time and right place.

Health system:

- Increased focus and clarity of required action on prevention of cardiovascular diseases.
- Better co-ordination of CVD care across the entire disease cycle.

Nationally:

- Alignment of policies and initiatives that support prevention of risk factors.

Strategic Pillar 2: Patient empowerment

Cardiovascular disease and its risk factors such as diabetes are chronic conditions which last and progress over years and decades. Hence, taking appropriate medication in the right way and the right time needs to become part of routine lifestyle for patients. In addition to compliance with medication, a significant

part of treatment involves hands-on management such as measuring blood sugar and blood pressure, modifying meals and physical activity. These need to be handled by patients and their carers in home setting. Therefore, the key to prevent development of complications and medical emergencies is that patients and their carers are supported and empowered to self-manage these chronic conditions.

A range of programs for patient education and empowerment are being piloted or in some instances routinely offered by various acute and primary care providers in Qatar. These interventions are delivered via a combination of face to face educational programs for e.g. diabetes clinics and dietary counselling by HMC and reading materials. However, these programmes tend to be varied, optional and not always accessible to patients in a systematic manner.

When initially diagnosed:

1. 38% of patients reported that they received too little information
2. Only 16% received annual appointment and check-up plans
3. Only 15% received a healthcare checklist
4. Only 14% received information on what to expect (i.e. how to live with and manage their chronic condition after diagnosis)

There is an appetite from the population to use technology as an educational and self-care tool:

- 89% of people would like reminders sent to their phones
- 74% would access information for conditions like diabetes if it was available
- 69% would like to monitor their health electronically
- 52% would like to be able to text or email their doctors

Source: MoPH patient survey (SCH, 3-2015) reported in Qatar National Diabetes Strategy 2016-22

The patient empowerment strategic pillar will deliver the following:

1- **Make medical knowledge accessible for public**

Although a lot of content is available online these days, this information is not validated or relevant in Qatar context. Therefore, over the next 3-5 years, all relevant public health and, if required, clinical practice guidelines will be summarised, simplified and tailored to the needs of the specific population groups and translated for lay public consumption.

2- **Standardised national 'curriculum'**

A standardised national curriculum for CVD and risk factor management education will provide minimum standards for:

-Disease-specific education (disease life-cycle, management and follow-up, complications and warning signs, when to seek help, medications, diet management etc)

-Skill-transfer (insulin injections, blood sugar and blood pressure monitoring, self-help apps, preparing healthy meals, relevant physical activity training etc).

This will be co-developed with primary and acute care providers and uniformly available in diabetes/ CVD/ stroke/ heart clinics as well as in primary care clinics. The providers will be able to publish the information under their own brand and add any additional content, of required but with permission from MoPH to ensure alignment and avoid confusion for patients.

3- **Self-management plans and care plans**

To ensure that the patients are empowered to take part in their care decisions and understand what is required from them and their carers in helping managing their condition, all care providers will be required to develop joint annual self-management plans and care plans with patients to reduce their risk (physical inactivity, smoking, unhealthy diet) of developing CVD or managing their intermediary health conditions such as diabetes and hypertension.

4- **Chronic disease self-management program**

All newly diagnosed patients with complex CVD conditions such as diabetes, heart diseases or stroke will be offered chronic disease self-management training (CDST) and self-care tool kit. This will be a community-based patient-centred, educational program, based on self-efficacy theory that aims to empower participants to develop necessary skills for management of medical, social and emotional factors related to chronic long-term conditions. This program will offer in person and online workshops that involve weekly sessions over 6-8 consecutive weeks on topics such as healthy eating, exercise, cognitive symptom management, measuring blood sugar levels etc, medication management, goal-setting, and action planning.

5- **Multi-channel approach for effective education and empowerment**

The chronic disease management program and other educational opportunities will be provided through tailored one-on-one sessions, group workshops and talks as well as made available online in an interactive format. In some GCC countries like Kuwait, a significant amount of this outreach is achieved through popular TV shows involving leading clinicians from the country. This approach will be explored along with community-outreach initiatives such as diabetes forums in mosques, schools and workplaces.

The healthcare providers will be encouraged to develop online telehealth platforms and applications for patients to track their appointments, parameters such as blood tests (HbA1C, lipid profile, renal function etc), track and manage multiple clinic appointments, send queries to their healthcare teams.

✓ Impact and outcomes of this strategic pillar are expected to be as follows:

Patients

- Understand the evolution of disease and their role in the care plan.
- Take charge of co-managing their condition with their healthcare teams.
- Improved quality of life and longer life span.

Health system

- Better patient compliance with medication and care plans
- Standardised communication to patients who move across providers.
- Reduced severity and overall burden of cardiovascular diseases on the health system

Nationally

- Reduced healthcare spending and increased economic productivity of residents diagnosed with CVD.

Strategic Pillar 3: Care delivery

Cardiovascular diseases are progressive conditions and their management changes as the disease progresses. During the lifespan, patients develop secondary complications as well as other short-term and long-term illnesses.
Effective management requires a multidisciplinary involving multiple health care professionals, including primary health care and specialist doctors, and a vast array of allied health professionals such as diabetes

educators, nutritionists, podiatrists etc. As a result, patients end up having to navigate a myriad of health care appointments and follow-ups for management of their disease.

A well-integrated and seamlessly organised care delivery system is critical for (secondary) prevention of CVD such that patients with diabetes or high blood pressure do not suffer from stroke, heart attacks and severe complications such as limb amputations and blindness.

Currently, the health system is fragmented between various providers of primary and secondary care. The care is often adhoc and patients are able to choose to see specialists directly. There are variable annual follow-up and review plans.
An in-depth healthcare professional survey (SCH, 4-2015) revealed that healthcare professionals in Qatar face challenges in guiding patients to access the right care at the right place at the right time. Only 58% of facilities provide formal training to staff in local and accredited international guidelines.

This pillar will aim to deliver the following initiatives towards improving the care delivery aspects of CVD management and secondary prevention:

1- **New model of care**

The new model of care will need to be jointly developed and agreed with the providers (Figure 10). It will aim to deliver the following:

a. Evidence–based standardised patient pathways:
System-wide patient pathways will be co-developed, agreed and implemented with the providers to screen, educate, track and treat people with cardiovascular diseases. MoPH clinical guidelines will be implemented as the minimum standard for clinical decision-making, through mandated responsibility of healthcare organisations and professionals.

b. Care-coordinator approach from initial screening onwards:
As the majority of routine management of CVD risk factors, including diabetes and hypertension, is managed in primary care, the primary care providers will play the pivotal role of coordinating patient care with appropriate follow-ups, referral to specialists and support services such as dietitians, smoking cessation, annual and quarterly investigation panels (HbA1C, blood pressure, BMI, lipids, foot and eye exam etc).

c. Clarity of roles and hand-over between primary and secondary care
To ensure that the patients receive the specialist input in their treatment when required, there will be clarity on roles and step-up and step-down referral guidelines and joint-care arrangements.

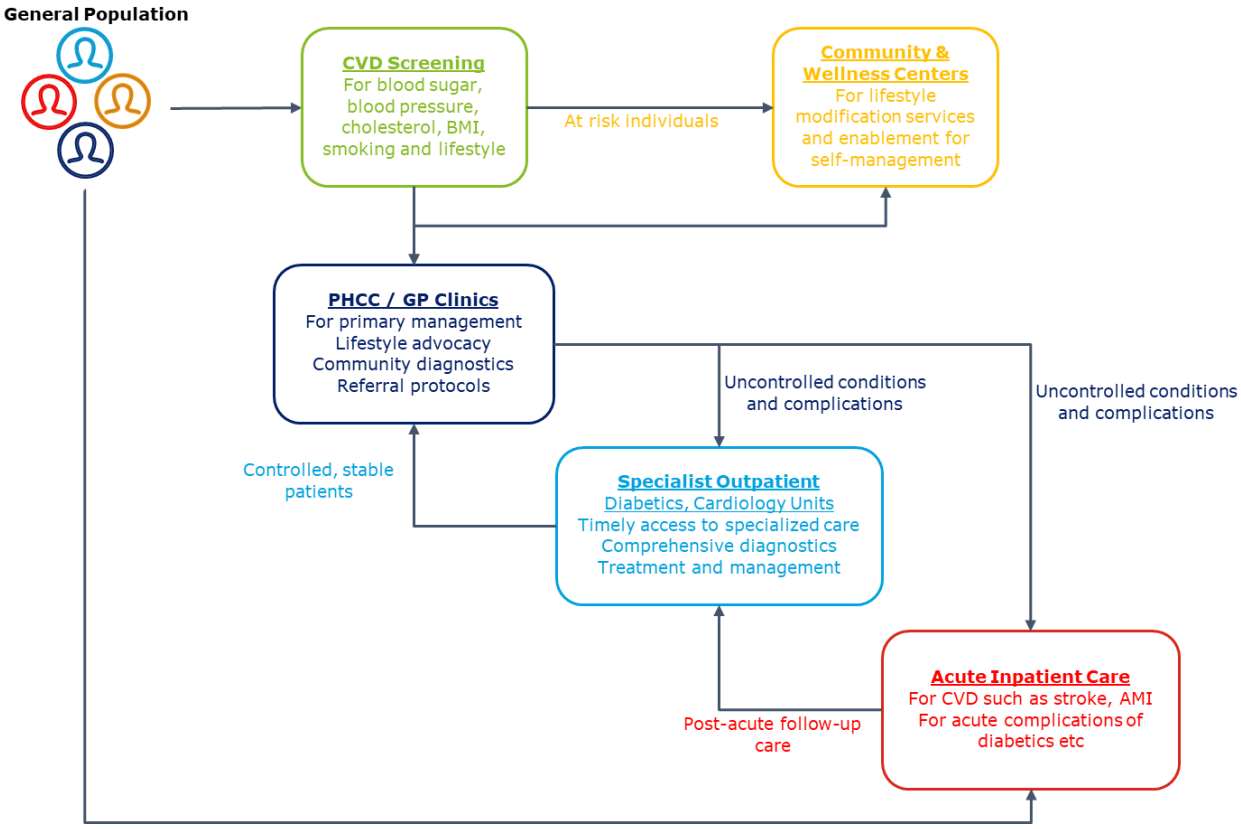


Figure 10: New model for delivering co-ordinated cardiovascular disease prevention and management services across the care-continuum

d. Community wellness centres/clinics
One-stop wellness support clinics will be established either within the primary care premises or in more accessible community settings to provide motivation, support and education for positive lifestyle changes such as weight-loss and smoking cessation and for providing training to patients and their carers for self-management of diabetes etc.

Interventions	Cross-cutting	1. Health screening-systematic detection of high BP, high blood sugar, high cholesterol, CVD risk 2. Health plan: system level action to support guideline implementation and standardization pathways 3. Support for patient empowerment, individual behavior changes and self-management				
	High BP detection & treatment	Diabetes* prevention	Diabetes Management	CVD lifestyle risk intervention	CKD** detection & management	Atrial fibrillation & anticoagulation
Opportunities	X undiagnosed. 40% poorly controlled	X with NDH. Most do not receive intervention	X undiagnosed. More than half do not receive all core care processes	85% of familial hypocholesteremia is undiagnosed. Most people at CVD risk do not receive statins or active lifestyle interventions	Mostly undiagnosed is non-systematic patients. Many have poor BP & proteinuria control	30% undiagnosed. Over half untreated or poorly controlled
Evidence	BP lowering prevents strokes and heart attacks	Intensive behavior changes reduces T2DM risk 30-60%	Control of BP, Hb1Ac and lipids improves CVF outcomes	Behaviour change and statins reduce lifetime risk of CVD	Control of BP, CVD risk and proteinuria improves outcomes	Anticoagulation prevents 2/3 of strokes in AF
Risk Conditions	Hypertension accounts for 50% of all strokes & heart attacks, plus CKD & dementia	Non-diabetics hyperglycemia (Pro-diabetes) leads to marked increase in Type 2 DM and CVD at an earlier age	Diabetes leads to increased risk of stroke, heart attacks, chronic kidney disease, limb amputation and blindness	High risk of stroke, heart attack, cancers and many other chronic disease	Chronic kidney disease leading to kidney failure and premature death	Atrial fibrillation increase the risk of CVD by 5 fold.
CO-ORDINATED PRIMARY CARE DELIVERY						
Target Outcome	Reduction in premature mortality from stroke & heart attack and kidney failure Reversal of cardiovascular risk Reduction in diabetes related limb amputations and blindness					

* Type 2 Diabetes, ** Chronic kidney disease

Figure 11: CVD Health Plan: Summary of risk detection and management in Primary Care

2- **Manpower development and training to support new model of care**

a. To support the new care model, all healthcare professionals will be trained on the basics of CVD risk factor education, screening, recording information and patient pathways. By 2022, most doctors, nurses, educators, and other healthcare professionals will check CVD risk factor status as standard and will be able to offer screening immediately if required. (refer strategic pillar 4)

b. To ensure an integrated team approach, besides clarity of roles, the following new roles will be created who will work closely with the primary care teams:
Community-based Educators: who will facilitate patient empowerment, coach through lifestyle interventions, and help patients take full control of their health.

Community champions: trained volunteers who may be other patients, caregivers, social workers etc.

3- **Linking of patient records to ensure continuity of care**

✓ Impact and outcomes of this strategic pillar are expected to be as follows:

Patients

- Single point of access for care
- Clarity on treatment plans and follow-up required
- Consistent high quality and continuity of care at all times, at all sites

Health system

- Clear accountability to patients
- Clarity on treatment plans and where to send patients next (referral pathways)
- Improved productivity

Nationally

- World-class healthcare system

Strategic Pillar 4: Human capability and capacity building

An in-depth healthcare professional survey (SCH, 3-2015 ,4-2015) revealed that:
62% of facilities struggled to recruit appropriate levels of staff
46% of facilities reported they have no clear method of recruitment

Source: Qatar National Diabetes Strategy 2016-22

Effective management of CVD risk factors requires a multi-disciplinary team involving multiple health care professionals. The role of the allied health professionals such as nutritionists, podiatrists and health educators is as vital as that of doctors. It is reported that there are gaps in numbers and appropriate training for staff to perform.

To overcome the gaps in CVD risk factor prevention and management, it is essential build-up a well-trained public health and clinical staff.

This strategic pillar would be delivered in close partnership with MoPH Human Ressource Department and the human resource leads of acute and primary care providers. The key initiatives are as follows:

1- **Skill mix and demand projection**

Public health, community and primary care and support workforce model will be developed to determine what the right workforce mix looks like to deliver the new model of care. This will also be used to project the number of staff needed by sites and roles.

2- **Competency framework and training**

A common core competencies framework for public health, health promotion and NCD management will be developed for clinical and non-clinical staff.

A diabetes and CVD management accreditation programme will be rolled out in partnership with academic institutions to implement training curriculum for existing staff in form of continuous education for all core professional groups and also as part of relevant graduate and undergraduate medical, allied health and nursing education programs.

3- **Strengthen recruitment & retention**

Additional manpower will be made available in time to support the delivery of the new care model, through:

- targeted overseas recruitment
- collaboration with universities and other educational organizations in Qatar
- partnerships with sourcing organizations

A series of regular common case management forums, technical updates and scientific conferences will be introduced with multidisciplinary attendance across all providers and public health professionals. This will help foster a learning community amongst the staff and help team working and staff retention.

In addition, formal performance targets, appraisal and career path across the public health manpower as an initiative under the national public health strategy will help improve productivity and effective deployment of existing manpower involved in CVD prevention.

4- **Formally develop and engage volunteers and third sector agencies**

Self- management and health education certification programs (online and classroom) for community volunteers and other educators will be developed with higher-education partners. In addition, they will be invited to relevant multi-disciplinary forums to help develop confidence and improve joined-up working.

✓ Impact and outcomes of this strategic pillar are expected to be as follows:

Patients
- Timely appointments
- High quality care and better care experience in healthcare and community settings

Health system
- Better staff satisfaction
- Improved retention rates
- Improved delivery of care

Nationally
- World-class healthcare system

Strategic Pillar 5: Information Management

Effective management of chronic diseases requires patients to be followed by different healthcare professionals at different facilities and at different times. Sharing patient information is key for coordinated care. Additionally, providing patients access to their own health information supports empowerment and self-care.

All healthcare establishments in Qatar have electronic patient record systems at varying scales of implementation. However, there are challenges in exchange of information across organisations.

Information technology infrastructure is key to the delivery of the new integrated care model and strengthening public health surveillance for cardiovascular diseases. This strategic pillar will drive the following:

1- **Improvement in data linkage for cardiovascular disease for continuity of care**

- National data dictionary and minimum dataset for diabetes, stroke and myocardial infarction.
- Data sharing agreements between primary and acute care providers.
- Standardised referral (primary care to specialists) and discharge form (specialist outpatient/ hospitals to primary care) s for CVD patients
- Enhanced electronic connectivity and information sharing between PHCC, HMC and other providers with the ultimate aim of a singled national electronic health record for each Qatar resident across the spectrum of care, as part of the nationwide eHealth Strategy.

2- **Establish national registries**

Disease specific patient- level databases/ registers will be developed for key conditions provide valuable data for sentinel surveillance and monitoring impact of healthcare interventions at population level. These databases will also provide invaluable data for research of public health importance.
The first registry to be developed will be that for diabetes. This will be patient- level data collected and integrated from screening as well as care provider databases into a central registration system and made available in anonymised form for tracking disease trends, impact of prevention and treatment initiatives and for research.

Initial phase of data collection will start with a small set of pre-agreed data and depend significantly on pulling the requisite ifnormation from provider records (either digitally or

manually). Going forward, this will evolve as a data warehouse that will be automatically integrated with provider electronic patient record and laboratory systems across primary and secondary care.

3- **Strengthen public health analytics and intelligence for CVD**

This will include development of a dashboard for key non-communicable diseases include CVD with output and outcome indicator tracker mapped to initiatives under each of the strategic pillars. A 'community relationships and network' management system will be piloted with aim to support community outreach.

4- **Pilot smart applications for at-risk population and patients**

With multiple applications and wearable technology available to support lifestyle changes and self-management, MoPH will work with identified technology partners to pilot and validate and adapt the applications or develop new applications which are relevant in Qatar context.

Remote patient monitoring and management is quite well-established for managing complex chronic diseases. Under this pillar, appropriate telehealth / telemedicine solutions will be pilot for patients with complex needs especially those who are frail elderly, bed-ridden or institutionalised.

✓ Impact and outcomes of this strategic pillar are expected to be as follows:

Patients
- Digital access to medical information and healthcare professionals
- Continuity of care across different sites and services

Health system
- Improved access to patient data and care co-ordination
- Improved public health surveillance and informed decision making
- Availability of data for healthcare research

Nationally
- Global recognition for advanced and effective adoption of technology in healthcare.

Strategic Pillar 6: Research

The health and life science research sector in Qatar has been steadily growing over the past few years. Access to invaluable healthcare genotypic and phenotypic data representative of Middle-eastern population, advancement of information technology and regulatory policies provides an excellent opportunity to advance the understanding of genomic and social determinants of cardiovascular diseases and underlying metabolic disorders.

Key initiatives under this strategic pillar to achieve a more concerted effort at the national level to drive research in this area of public health priority are as follows:

1- **Integration of data to support research**

- To enable quality research, data from multiple sources will be collated and made easily accessible to researchers. This research platform will be made available for both national and international consortia of collaborating researchers will enable analysis of anonymized patient data for clinical research.

2- **Prioritization on the national research agenda**

Diabetes, hypertension, heart and kidney disease are disproportionately distributed by ethnicity and geography in the local population. Hence, the key areas of CVD research of interest to Qatar are those that will help refine the current and future public health and care interventions. These include:

- Validation of risk calculators tools to predict undiagnosed CVD and diabetes in the local population,
- To understand the relationships and interdependencies among clinical phenotypes, human behaviour, and molecular, genomic and social determinants and their contribution to health disparities in cardiovascular disease.
- Participation in large international trials on epidemiology of cardiovascular diseases especially effects of social determinants on cardiovascular disease morbidity and mortality among disproportionately affected population subgroups such as Middle –east and South Asia.
- To adapt/ participate in development of molecular and genetic epidemiology models to identify patterns and determinants of common chronic diseases in middle-eastern populations.

3- **Strengthen research governance**

- Work with the Qatar National Research Fund (QNRF) and the National Research Committee to establish a robust governance process for developing and driving the public health research agenda. This could include:
    - A National Public Health Research Committee as the governing body for research in areas of public health priority starting with Diabetes and other common NCD risk factors.
    - Ring-fenced funding from QNRF for public health and health system research
    - Target grant calls in priority areas of public health importance, including pilots for new interventions.

4- **Embed formative evaluation research in public health programs**

- It is usually difficult to evaluate the impact of complex public health interventions. All health interventions implemented under the CVD strategy will have a formative research component embedded in developmental stage through use of qualitative and quantitative measures to provide invaluable insight into the impact of the interventions in the short to medium term.

✓ Impact and outcomes of this strategic pillar are expected to be as follows:

<u>Patients</u>
- Access to new treatments

<u>Health system</u>
- A culture of continuous evaluation and improvement in healthcare.
- Involvement is research builds excellence in the organisation and helps in attracting and retaining talented staff from all over the world.

<u>Nationally</u>
- Achieving the national ambition of becoming a leading life science research and development

5.5 IMPLEMENTATION PLAN

The implementation of CVD risk factor prevention strategy is complex involving accountability of delivery across multiple internal healthcare and multi-agency stakeholders in public, social and private sector. The following implementation plan is based on the review of critical success factors behind successful public health implementation[34].

5.5.1 Governance

The day-to-day programme governance will be coordinated by the NonCommunicable Disease division of Ministry of Public Health under the oversight of Public Health Strategy Implementation group.

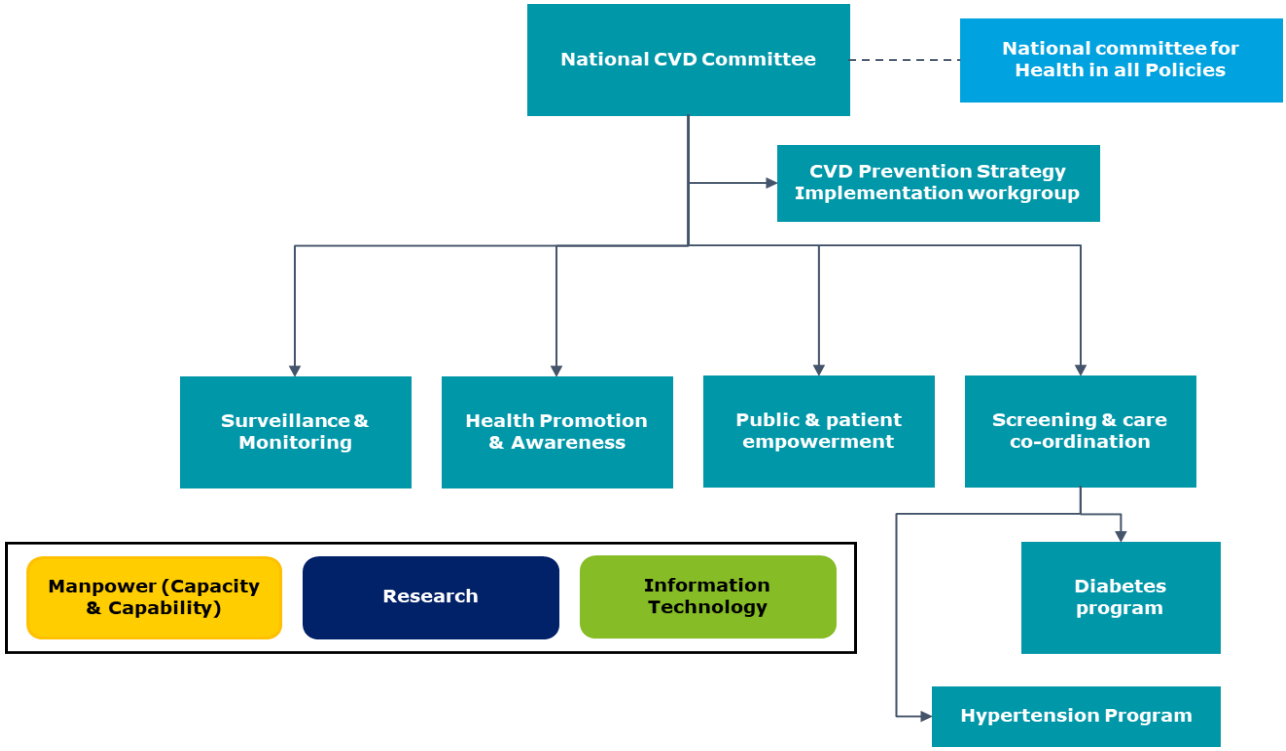


Figure 12: Governance Structure for Implementation of a National CVD Prevention Strategy

The National CVD committee should include nominated representatives from key primary and acute healthcare organisations as well as the key stakeholders identified in Table 3. This is to lay the foundation for a whole system approach towards creating supporting environment and policies conducive to cardiovascular disease prevention at national level.

Ministries	Areas of Accountability
Industry, Labour & Social affairs	Workplace health promotion including health screening for NCD and lifestyle advocacy. Pro-health work environment and culture etc. Leading community involvement and action for health promotion and healthy lifestyle with religious and non-governmental organisations.
Economy & Commerce	Healthy food imports and production; accessible food labelling etc.
Justice	Health friendly regulations such as tobacco control, extended food safety regulations, trans fat and sugar tax etc.
Education	Health promoting schools and higher education including health education and healthy living embedded into daily educational life etc.
Infrastructure Planning, Municipality & Environment	Healthy and safe environments that promote active lifestyle e.g. community gyms, indoor walking trails, pedestrian walkways and crossings, green buildings etc.
Culture & Sports	Development of a culture of sport and active lifestyle across gender and age groups.

Table 3: Stakeholder in collaborative partnership for cardiovascular disease prevention.

5.5.2 Accountability, Monitoring & Reporting

It is proposed that each of the strategic pillars is overseen by a named senior officer within the MoPH team, the pillar-owner, who will drive the following:
- Development of detailed action plan for each initiative
- Identification with stakeholders involved and consensus on ownership of implementation
- Agreement on key delivery milestones
- Monitoring
- Periodic reporting to the NCD divison and national CVD committee.

The implementation of each strategic pillar would require working closely across the various directorates of MoPH, as well as other Ministries and healthcare providers. Therefore, in addition to a joint oversight by the National CVD Committee, each strategic pillar would need cross-functional working groups which will be coordinated by the pillar-owners to implement the agreed actions.

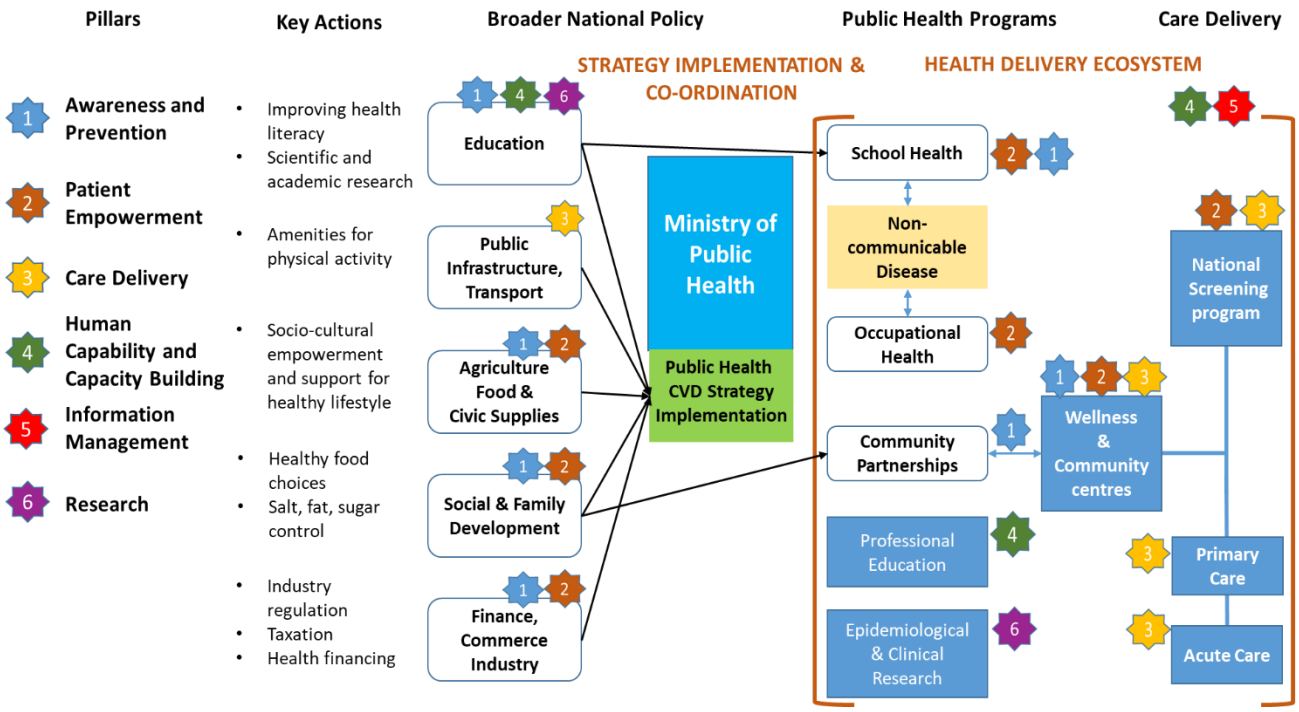


Figure 13: Cross-functional Working groups across 6 strategic pillars

5.5.3 Communication and Engagement

CVD prevention strategy implementation will be complex programme of work requiring continuous engagement and tracking across multiple stakeholders. Therefore, a communication and engagement plan is a key aspect of implementation. The key featutes of this plan would include the following:

1. Cross-sector CVD strategy sign-off and briefing seminar
2. Dedicated implementation tracker and quarterly update report by the project team to all stakeholders
3. Annual CVD coordination and planning session to:
   - Agree on the key annual priorities for delivery under CVD strategy
   - Identify key policies which can impact common NCD risk factors and healthy lifestyle in general within each Ministry's work plans
   - Progress updates CVD actions and agreement on next steps

5.5.4 Key deliverables & Timeline

A set of key performance indicators (KPIs) will have to be developed, along with a detailed implementation plan, to illustrate proposed success measures for the implementation of the National Cardiovascular Disease Risk Factors Prevention Strategy. However, these indicators will be further defined and refined based on agreed priorities for implementation.

An example KPI stated in the PHS 2017-2022, would be: Percentage of adults, aged 18 to 64 years) who areoverweight or obese (BMI greater or equal to 25 kg/m^2).

A draft of the implementation plan timeline has been included in Annex E. The implementation plan timeline will be discussed and drafted with inputs of NCD Division as it will be closely linked to divisional and MoPH strategic priorities, other co-dependencies and delivery capacity.

This strategy forms part of Qatar's overarching National Health Strategy and closely aligned to wider NHS projects as described in Annex A and B.

5.5.5 Pilot to scale framework

Scaling up complex health interventions to population-wide level needs guided efforts. The implementation of the CVD risk factor strategy would use the WHO scale-up framework[35] to ensure that the all-successful pilot interventions are implemented in a timely and planned manner. The details of this framework are included in Annex F.

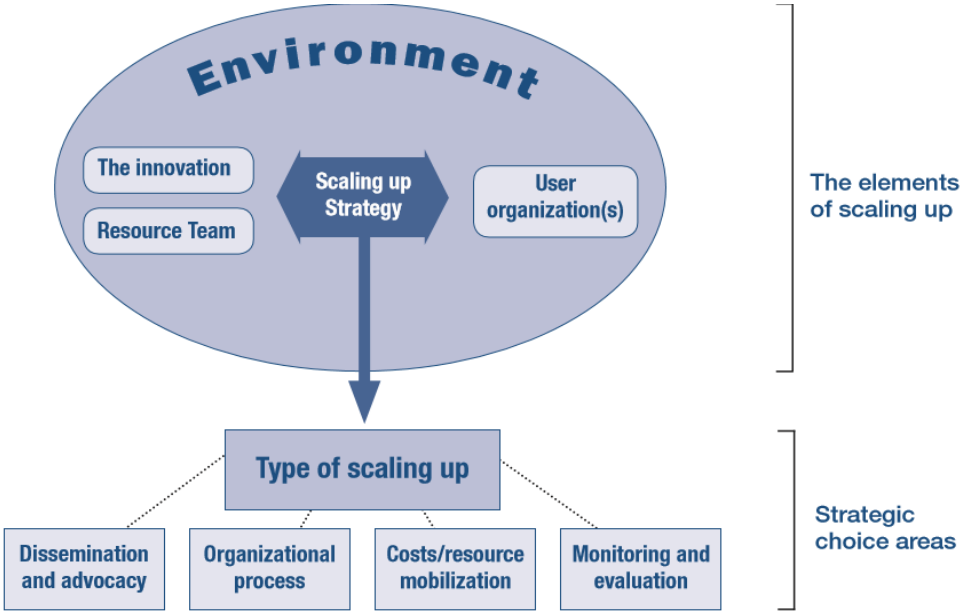


Figure 13: The ExpandNet/WHO framework for scaling up, WHO/ExpandNet, Geneva, 2011

6 BIBLIOGRAPHY

1. Yusuf, S., Hawken, S., Ôunpuu, S., Dans, T., Avezum, A., & Lanas, F. et al. (2004). Effect of potentially modifiable risk factors associated with myocardial infarction in 52 countries (the INTERHEART study): case-control study. *The Lancet*, *364*(9438), 937-952.
2. WHO. (2019). Noncommunicable diseases. Retrieved from https://www.who.int/en/news-room/fact-sheets/detail/noncommunicable-diseases
3. Roy, A., Rawal, I., Jabbour, S., & Prabhakaran, D. (2017). Tobacco and Cardiovascular Disease: A Summary of Evidence. *Disease Control Priorities*.
4. National health Strategy, MoPH 2018-2022
5. Christos, P., Chemaitelly, H., Abu-Raddad, L., Gehani, A., Deleu, D., & Mushlin, A. (2013). Prevention during the epidemiologic shift to chronic illness: a case control study of risk factors associated with cardiovascular disease in Qatar. *Journal Of Local And Global Health Perspectives*, (2013), 4.
6. Sidney S, Quesenberry CP, Jaffe MG, et al. Recent Trends in Cardiovascular Mortality in the United States and Public Health Goals. *JAMA Cardiol.* 2016;1(5):594–599. doi:10.1001/jamacardio.2016.1326
7. Health Report on the Nation of Qatar, July 2015. Johns Hopkins Bloomberg School of Public Health. (n.d.).
8. Haj Bakri A and Al-Thani A. Chronic Disease Risk Factor Surveillance: Qatar STEPS Report 2012. The Supreme Council of Health. Qatar. 2013.
9. Bener, A. (2006). Prevalence of Obesity, Overweight, and Underweight in Qatari Adolescents. *Food And Nutrition Bulletin*, *27*(1), 39-45.
10. Rootwelt, Medical Student, C., Beinnes Fosse, K., Tuffaha, A., Said, H., Sandridge, A., Janahi, I., Greer, W. & Hedin, L. (2014, November). Qatar´ s Youth Is Putting On Weight: The Increase In Obesity Between 2003 And 2009. In *Qatar Foundation Annual Research Conference* (No. 1, p. HBSP1130).
11. Qatar National Census 2010; Ministry of Development Planning and Statistics; https://www.mdps.gov.qa/en/statistics1/StatisticsSite/Census/census2010/Pages/publications.aspx
12. Impact of population ageing on the costs of hospitalisations for cardiovascular disease: a population-based data linkage study." BMC health services research 14
13. Chahine, G., Bitar, J., Assouad, P., & Abi Chaker, S. (2013). The $68 billion challenge, quantifying and tackling the burden of chronic diseases in the GCC.
14. Prevention of cardiovascular disease : guidelines for assessment and management of total cardiovascular risk.©WHO 2007
15. Lim, S. S., Vos, T., Flaxman, A. D., Danaei, G., Shibuya, K., Adair-Rohani, H., & Aryee, M. (2012). A comparative risk assessment of burden of disease and injury attributable to 67 risk factors and risk factor clusters in 21 regions, 1990–2010: a systematic analysis for the Global Burden of Disease Study 2010. *The lancet*, *380* (9859), 2224-2260.
16. Jackson, R., Lynch, J., & Harper, S. (2006). Preventing coronary heart disease: Does Rose's population prevention axiom still apply in the 21st century?. *BMJ: British Medical Journal*, *332*(7542), 617. West of Scotland Coronary Prevention Group. (1996). West of Scotland Coronary Prevention Study: identification of high-risk groups and comparison with other cardiovascular intervention trials. *The Lancet*, *348*(9038), 1339-1342.
17. Ferrario, M., Chiodini, P., Chambless, L., Cesana, G., Vanuzzo, D., & Panico, S. et al. (2005). Prediction of coronary events in a low incidence population. Assessing accuracy of the CUORE Cohort Study prediction equation. *International Journal Of Epidemiology*, *34*(2), 413-421.
18. Jackson, R. (2000). Updated New Zealand cardiovascular disease risk-benefit prediction guide. *BMJ*, *320*(7236), 709-710.

19. Assmann, Gerd & Cullen, Paul & Schulte, Hagen. (2002). Simple Scoring Scheme for Calculating the Risk of Acute Coronary Events Based on the 10-Year Follow-Up of the Prospective Cardiovascular Munster (PROCAM) Study. Circulation. 105. 310-5.
20. Conroy, R. (2003). Estimation of ten-year risk of fatal cardiovascular disease in Europe: the SCORE project. *European Heart Journal*, *24*(11), 987-1003.
21. Sheridan, S., Pignone, M., & Mulrow, C. (2003). Framingham-based tools to calculate the global risk of coronary heart disease: a systematic review of tools for clinicians. *Journal of general internal medicine*, *18*(12), 1039-1052.
22. Liu, J. (2004). Predictive Value for the Chinese Population of the Framingham CHD Risk Assessment Tool Compared With the Chinese Multi-provincial Cohort Study. *JAMA*, *291*(21), 2591.
23. Jee, S. H., Jang, Y., Oh, D. J., Oh, B. H., Lee, S. H., Park, S. W., ... & Yun, Y. D. (2014). A coronary heart disease prediction model: the Korean Heart Study. *BMJ open*, *4*(5), e005025.
24. Lin, F. J., Tseng, W. K., Yin, W. H., Yeh, H. I., Chen, J. W., & Wu, C. C. (2017). Residual Risk Factors to Predict Major Adverse Cardiovascular Events in Atherosclerotic Cardiovascular Disease Patients with and without Diabetes Mellitus. *Scientific reports*, *7*(1), 9179.
25. Lam, T. H., He, Y., Shi, Q. L., Huang, J. Y., Zhang, F., Wan, Z. H., ... & Li, L. S. (2002). Smoking, quitting, and mortality in a Chinese cohort of retired men. *Annals of epidemiology*, *12*(5), 316-320.
26. Glynn, T. J., Shopland, D. R., Manley, M., Lynn, W. R., Freedman, L. S., Green, S. B., ... & Chapelsky, D. A. (1995). Community Intervention Trial for Smoking Cessation (COMMIT): II. Changes in adult cigarette smoking prevalence. *American Journal of Public Health*, *85*(2), 193-200.
27. Marmot, M., Allen, J., Goldblatt, P., Boyce, T., McNeish, D., & Grady, M. (2010). Fair society, healthy lives. *The Marmot Review*, *14*. [1]
28. Franek J. Self-management support interventions for persons with chronic disease: an evidence-based analysis. *Ont Health Technol Assess Ser. 2013;13(9)*:1-60. Published 2013 Sep 1.
29. Fu Dangbao, Fu Hua, Patrick McGowan et al Implementation and quantitative evaluation of chronic disease self-management programme in Shanghai, China: randomized controlled trial. *Bulletin of the World Health Organization 2003*, 81 (3)
30. Namrata H, Xinyue F, Weifen L, Soong YK. Evaluation of a Chronic Disease Self-Management Program in Singapore. *Ann Chron Dise. 2016; 1(1)*: 1001.
31. Horton, R. (2013). Non-communicable diseases: 2015 to 2025. *The Lancet*, *381*(9866), 509-510.
32. WHO. (2019). WHO | What is a health promoting school?. Retrieved from https://www.who.int/school_youth_health/gshi/hps/en/.
33. WHO | Workplace health promotion. (2019). | The workplace: A priority setting for health promotion. Retrieved from https://www.who.int/occupational_health/topics/workplace/en/
34. Frieden, T. R. (2014). Six components necessary for effective public health program implementation. *American journal of public health*, *104*(1), 17-22
35. World Health Organization. (2010). Nine steps for developing a scaling-up strategy

7 APPENDICES

Annex A: Annual surveys in different health areas

Annex B: List of initiatives from NCDs, Healthy Lifestyle, Reduced salt and fat in diet and Tobacco Control.

Annex C: The terms of reference and membership of the National Diabetes Committee

Annex D: List of CVD-related Guidelines

Annex E: Implementation plan timeline template

Annex F: WHO/ExpandNet framework for scaling up

7.1 Appendix A

List of surveys mapped to strategic initiative 2.2.2, Qatar Public Health Strategy 2017-2022

Health Area Objective	Sub-objective	Project
CVD1: Reduce the prevalence and burden associated with cardiovascular disease through promotion of healthy behaviours including education programs at primary care centers	Increase the % of age standardized prevalence of adults limiting their dietary salt	National survey on the risk factors every 5 years
O2: Implement targeted population-based oral disease prevention and promotion programs including healthy eating and tobacco cessation initiatives	O 2.1: Increase the % of Public Awareness on Risks of Unhealthy Diet on Oral Health	National Survey to determine the Baseline
O3: Implement annual oral and dental check-up for adults as per international recommendations	O 3.1: Increase the % of adults receiving oral screening	National Survey to determine Baseline
V1: Comprehensive and effective eye health education and screening services for the population	V1.1: Decrease the % of people living with blindness	National Survey to update the Baseline
	V1.2: Increase the % of registered diabetic patients who have an annual routine retinal examination	National survey to determine the baseline
	V1.3: Increase % of registered glaucoma patients being followed annually	National survey to determine the baseline
TC1: Establish a tobacco surveillance system (at a national level involving the primary care system) and conduct regular surveys as per GTSS	Conduct Survey every 5 years	Global Youth Tobacco Survey Global Adults Tobacco Survey (STEP)
MS1: Establish baselines and implement ongoing data collection on arthritis & musculoskeletal disorders	Increase awareness about Osteoarthritis (OA)	Survey for Osteoarthritis
	Decrease the % of OA in the population	Tendinitis Survey
	Decrease the % of Tendinitis	Osteoporosis Survey
MC3: Establish a preconception service within primary care	Increase the % of women of reproductive age who have reproductive planning & benefit from preconception care	Survey to Determine baseline
	Decrease the % of unplanned pregnancy	Survey to determine unplanned pregnancy
MC4: Implement initiatives to encourage and promote breastfeeding	Increase the % of mothers who breast-feed their newborns	Survey to determine baseline
C1: Implement national cancer myth refutation campaigns	Increase the number of national cancer myth refutation campaigns	Cancer Awareness Measure CAM
MH1: Raise public awareness about mental health and reduce the stigma associated with mental illness, including an increase in the Mental Health Attitudes and Awareness Index	Increase Mental Health Attitudes and Awareness Index (QMHAAI) which encompasses four measures: communication, awareness, stigma, services	Qatar Mental Health Attitudes and Awareness Baseline Survey every 2 years (2018, 2020, 2022)
CDC1: Implement and sustain effective HIV prevention programs, including education and awareness, and effective anti-retroviral therapy	CDC 1.3: Increase percentage of People who achieve viral load suppression.	Integrated bio-behavioral surveys (IBBS)

CDC6: Achieve and sustain recommended routine and adult immunization coverage for the whole population, including high risk groups	CDC6.1 Increase vaccine coverage for children	National vaccination coverage survey for comprehensive data collection and analysis
FS 1: Strengthen public awareness of food hazards and health risks associated with the consumption of unsafe food through the use of various effective forms of modern mass media communication channels	FS 1.1 Increase % of population aware of food safety best practices	Public Awareness Surveys
EH 1: Assess and achieve health- based air pollution standards and reduce risk from indoor and ambient air contaminants	EH 1.2: Increase the % of the population awareness about Asthma related to dust storms	Survey to determine baseline
	EH1.3: Increase the % of primary schools with healthy Indoor Air Quality IAQ	Survey to determine the baseline
RD1: Enhance the integrated asthma program in schools, including education regarding asthma awareness and support program	RD1.3: Increase the numbers of conducted surveyed Independent primary schools.	Asthma Friendly schools program Surveys

7.2 Appendix B

Healthy Lifestyle under the Public Health Strategy

Objective	Project
HL 1: Reduce the risk factors for chronic non-communicable diseases (unhealthy nutrition and physical inactivity)	Baby Friendly Hospital Initiative BFHI
	Breastfeeding Campaign
HL 2: Increase the overall public awareness of the positive health effects of healthy nutrition and engaging in regular physical activity to enable the population to easily make healthy nutritional choices	National Health Promoting Schools Programme
HL 3: Formulate and promote culturally appropriate and sustainable policies and legislations aimed at promoting food diversity, healthy eating habits and increasing physical activity in the population	Legislations & Policies Formulation
HL 4: Increase regular physical activity across the population with particular emphasis on schools, workplaces and targeted communities	Engagement and coordination framework
HL 5: Establishment of wellness services and health coaches into primary healthcare centers	Training Workshops for healthcare professionals
HL 6: Enhance collaboration with the private sector to promote the importation, production and distribution of food products which contribute to a healthier and more balanced diet	Fat, Salt and Sugar Reduction Initiative

Tobacco Control under the Public Health Strategy

Objective	Project
TC1: Establish a tobacco surveillance system (at a national level involving the primary care system) and conduct regular surveys as per GTSS	Survey
TC2: Implement a tobacco law enforcement framework that is aligned with the WHO Framework Convention on Tobacco Control compatible policy and legislation	Tobacco law Implementation
	Comprehensive tobacco e-system for facilities requests
	Tobacco free CSR initiative
	Staff CoI Declaration
	GHW implementation
	Functioning user friendly public communication system
TC3: Establish standardized, quality, integrated and accessible tobacco cessation services including a national quit line and website, availability of all options of nicotine replacement therapy, and support through PHCC	Quit line
	Website linked to services
	Awareness campaign
	An integrated standardized quality cessation services
TC4: Establish a functional comprehensive tobacco product taxation model that includes customs and excise taxes	Taxation system for tobacco products
	Ratification of the protocol on illicit tobacco trade

7.3 Appendix C

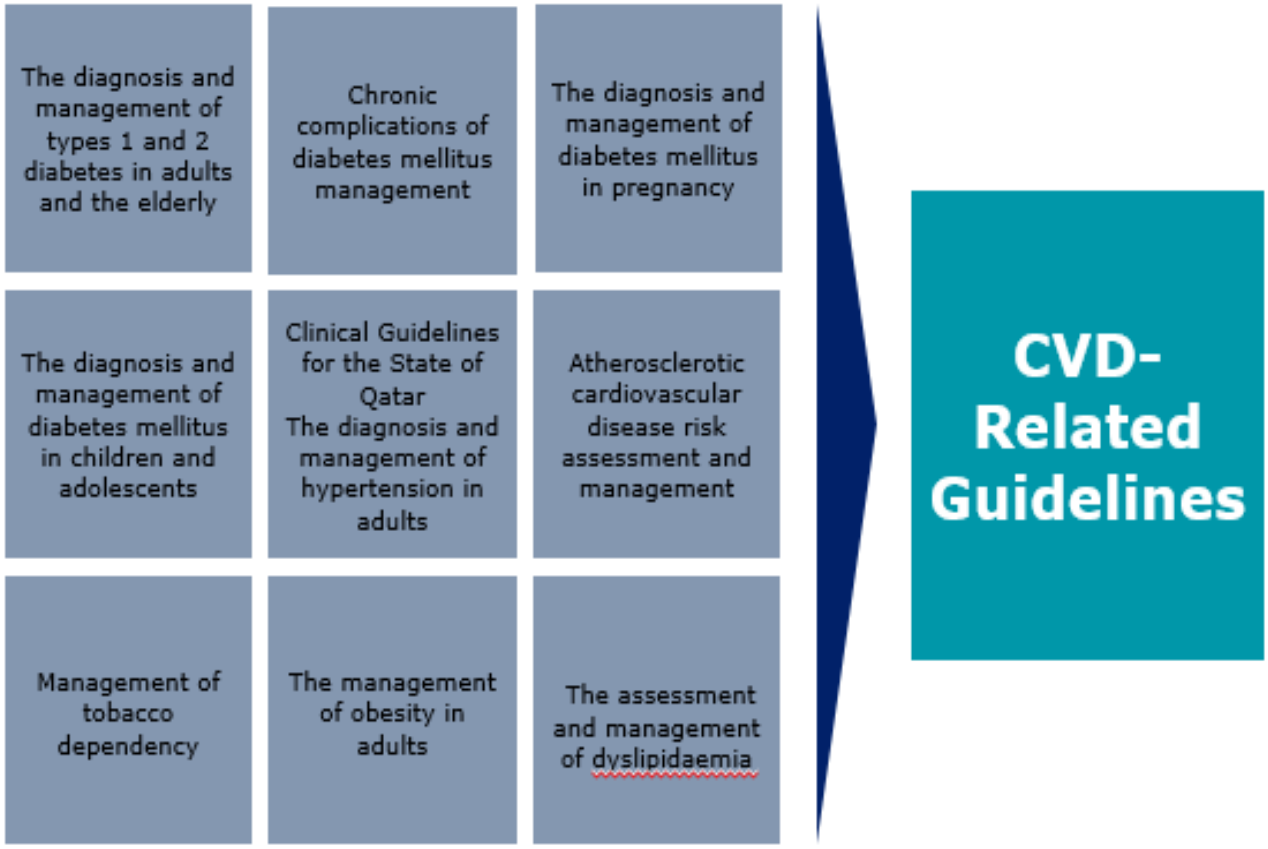
National Diabetes Committee Members

Name	Organization
Professor Abdul Badi Abou-Samra	HMC
Dr. Al-Anoud Al-Thani	MOPH
Dr. Aiman Hussein Farghaly	MOPH
Dr. Dr. Kholood Al Mutawa	MOPH
Dr. Shahrad Taheri	Weil Cornell Medical College
Dr. Abdulla Alhamaq	Qatar Diabetes Association
Dr. Abdulla Al Naama	PHCC
Dr. Mohsin Mesmar	PHCC
Dr. Fawziya Al Khalaf	Sidra
Dr. Dhabia Al-Muhannadi	HMC

7.4 Appendix D

List of CVD-related guidelines

A detailed description of all these guidelines can be found on the MoPH website



7.5 Appendix E:

Implementation Plan Timeline Template

To be discussed and drafted with inputs of NCD Division as the implementation timeline will be closely linked to divisional and MoPH strategic priorities, other co-dependencies and delivery capacity

Pillar I: Awareness and Prevention	2019		2020				2021				2022				2023				2024			
Initiatives	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1- Standardised and simplified CVD prevention content across national agencies and providers																						
2- Multipronged communication plan to improve dissemination of messages																						
3- Active community engagement for prevention																						
4- Translating awareness into action																						
5- Cardiovascular disease screening programme																						
6- Health plan																						
7- Tracking and linkage of data across the care pathway																						

Pillar II: Patient Empowerment	2019		2020				2021				2022				2023				2024			
Initiatives	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1- Make medical knowledge accessible for public																						
2- Standardised national 'curriculum'																						
3- Self-management plans and care plans																						
4- Chronic disease self-management program																						
5- Multi-channel approach for effective education and empowerment																						

Pillar III: Care Delivery	2019		2020				2021				2022				2023				2024			
Initiatives	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1- New model of care																						
2- Manpower development and training to support new model of care																						
3- Linking of patient records to ensure continuity of care																						

Pillar IV: Human Capability and Capacity Building	2019		2020				2021				2022				2023				2024			
Initiatives	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1- Skill mix and demand projection																						
2- Competency framework and training																						
3- Strengthen recruitment & retention																						
4- Formally develop and engage volunteers and third sector agencies																						

Pillar V: Information Management	2019		2020				2021				2022				2023				2024			
Initiatives	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1- Improvement in data linkage for cardiovascular disease for continuity of care																						
2- Establish national registries																						
3- Strengthen public health analytics and intelligence for CVD																						
4- Pilot smart applications for at-risk population and patients																						

Pillar VI: Research	2019		2020				2021				2022				2023				2024			
Initiatives	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1- Integration of data to support research																						
2- Prioritization on the national research agenda																						
3- Strengthen research governance																						
4- Embed formative evaluation research in public health programs																						

7.6 Appendix F

WHO/ExpandNet framework for scaling up, WHO/ExpandNet, Geneva, 2010

The nine steps for developing a scaling-up strategy

Attribute	Key questions	If necessary, find ways to:
Perceive need	1. Is there a perceived need for the innovation? Does the innovation respond to a policy priority? 2. Are there individuals within the organization who are advocates/champions of the innovation?	Strength perceived need/motivation through advocacy using both formal and informal channels. Identify and work advocates/champions for the innovation.
Implementation capacity	1. Does the user organization have capacity in: • Technical skills • Training • Logistics/supplies • Leadership/co-ordination • Monitoring/evaluation • Physical facilities and equipment • Values supportive of the innovation • Human resources • The policy and legal framework necessary to introduce the innovation? 2. Did the pilot project test ways to strengthen the capacity of the user organization? 3. Can the user organization absorb this innovation without negative impact on other programmes and services?	Begin expansion in areas where capacity is stronger. Identify opportunities for mobilizing/sharing resources within the user organization. Advocate for needed policy/legal change. Identify other ways of building necessary capacities during expansion. Test capacity strengthening before wide-scale expansion. How can these negative impacts be reduced or eliminated?
Timing and circumstances	1. Are there impending changes within the user organization that will affect scaling up? 2. Do these changes provide opportunities or constraints?	Adjust the scaling-up strategy to maximize opportunities and minimize constraints arising from these impending changes.

Strategic choice area	Key questions	If necessary, find ways to:
Dissemination/advocacy	1. What are appropriate approaches to advocate for the needed changes? 2. What formal and informal advocacy channels could be used? 3. How can a board sense of ownership be created so that the innovation will be viewed as part of routine programme operations	Strength perceived need/motivation through advocacy using both formal and informal channels. Identify and work advocates/champions for the innovation.
Organizational process	1. Who will be responsible for organizing the process of making the changes? 2. What technical support will be needed for the required changes and how can such support be mobilized? 3. Through what process can the various changes be initiated? 4. How can support for the innovation be linked to national health plans and health reform processes? 5. In what time periods should the changes be undertaken?	Begin expansion in areas where capacity is stronger. Identify opportunities for mobilizing/sharing resources within the user organization. Advocate for needed policy/legal change. Identify other ways of building necessary capacities during expansion. Test capacity strengthening before wide-scale expansion. How can these negative impacts be reduced or eliminated?
Cost/resource mobilization	1. Are resources for dissemination /advocacy and related costs of the necessary changes available? 2. How can budgetary processes be influenced and how can cost of required changes be inserted in national health budgets?	Adjust the scaling-up strategy to maximize opportunities and minimize constraints arising from these impending changes.
Monitoring and evaluation	1. How vertical scaling-up be monitored and evaluated? 2. What are appropriate indicators	Create simple ways of tracking activities and accomplishments related to vertical scaling-up and assess what else is needed. Use results from monitoring and evaluation to adjust the strategy as scaling-up proceeds.

8 GLOSSARY

CVD	Cardiovascular Disease
HMC	Hamad Medical Corporation
MI	Myocardial Infraction
NCD	Non-communicable disease
MoPH	Ministry of Public Health
PHCC	Primary Health Care Corporation
T2DM	Type 2 Diabetes Mellitus
WHO	World Health Organization

Deloitte.

This document has been prepared by the relevant legal entity for the sole purpose of providing a proposal to the parties to whom it is addressed in order that they may evaluate the capabilities of the relevant legal entity to supply the proposed services.

The information contained in this document has been compiled by the relevant legal entity and includes material which may have been obtained from information provided by various sources and discussions with your management which has not been verified or audited. This document also contains material proprietary to the relevant legal entity. Except in the general context of evaluating our capabilities, no reliance may be placed for any purposes whatsoever on the contents of this document or on its completeness. No representation or warranty, express or implied, is given and no responsibility or liability is or will be accepted by or on behalf of the relevant legal entity or by any of its partners, members, employees, agents or any other person as to the accuracy, completeness or correctness of the information contained in this document or any other oral information made available and any such liability is expressly disclaimed.

Other than as stated below, this document and its contents are confidential and prepared solely for your information, and may not be reproduced, redistributed or passed on, directly or indirectly, to any other person in whole or in part. Therefore you should not refer to or use our name or this document for any other purpose, disclose them or refer to them in any prospectus or other document, or make them available or communicate them to any other party. In any event, no other party is entitled to rely on our documentation for any purpose whatsoever and thus we accept no liability to any other party who is shown or gains access to this document.

This document is not an offer and is not intended to be contractually binding. Should this proposal be acceptable to you, and following the conclusion of our internal acceptance procedures, we would be pleased to discuss terms and conditions with you prior to our appointment.

Deloitte refers to one or more of Deloitte Touche Tohmatsu Limited, a UK private company limited by guarantee ("DTTL"), its network of member firms, and their related entities. DTTL and each of its member firms and their related entities are legally separate and independent entities. DTTL (also referred to as "Deloitte Global") does not provide services to clients. Please see www.deloitte.com/about to learn more about our global network of member firms.

Deloitte provides audit, consulting, financial advisory, risk advisory, tax and related services to public and private clients spanning multiple industries. Deloitte serves four out of five Fortune Global 500® companies through a globally connected network of member firms in more than 150 countries and territories bringing world-class capabilities, insights, and high- quality service to address clients' most complex business challenges. To learn more about how Deloitte's approximately 245,000 professionals make an impact that matters, please connect with us on Facebook, LinkedIn, or Twitter.

Deloitte & Touche (M.E.) is a member firm of Deloitte Touche Tohmatsu Limited (DTTL) and is a leading professional services firm established in the Middle East region with uninterrupted presence since 1926. DTME's presence in the Middle East region is established through its affiliated independent legal entities which are licensed to operate and to provide services under the applicable laws and regulations of the relevant country. DTME's affiliates and related entities cannot oblige each other and/or DTME, and when providing services, each affiliate and related entity engages directly and independently with its own clients and shall only be liable only for its own acts or omissions and not those of any other affiliate.

© 2018 (insert relevant legal name here). All rights reserved.